CONTRIBUTED PAPER

Building Resilient Web 3.0 Infrastructure With Quantum Information Technologies and Blockchain: An Ambilateral View

This article examines how state-of-the-art quantum information technologies are driving blockchain to open new horizons for the development of Web 3.0.

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ABSTRACT | Web 3.0 pursues the establishment of decentralized ecosystems through blockchain technologies, driving digital transformation in commerce and governance. With consensus algorithms and smart contracts grounded in cryptographic technologies, Web 3.0 enables secure and transparent digital services, such as digital identity, asset management, decentralized autonomous organizations (DAOs), and decentralized finance (DeFi), fostering integration between digital and physical economies. As quantum devices rapidly advance, Web 3.0 is being developed in parallel with the deployment of quantum cloud computing and quantum Internet. In this regard, quantum computing first disrupts the original cryptographic systems that protect data security while reshaping modern cryptography with enhanced quantum computing and communication capabilities. This article provides a comprehensive overview of blockchain-based Web 3.0, examining its quantum and postquantum advancements from two key perspectives. On the one hand, postquantum migration methods and quantum-resistant signatures offer robust solutions to safeguard blockchain against quantum threats. On the other hand, quantum and postquantum encryption and verification algorithms boost blockchain performance, creating a decentralized, secure, and value-driven system. Additionally, we outline potential applications of quantum blockchain and offer guidance for implementation within the Web 3.0 ecosystem. Finally, we discuss future directions for developing a provably secure and decentralized digital ecosystem.

KEYWORDS | Blockchain; decentralization; postquantum cryptograph; quantum computing.

Received 6 May 2024; revised 1 November 2024; accepted 11 December 2024. Date of publication 23 January 2025; date of current version 3 February 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB2902500; in part by the National Natural Science Foundation of China under Grant 62325203, Grant U22B2033, Grant 62072332, and Grant F2024201070; in part by the Program for Youth Innovative Research Team of Beijing University of Posts and Telecommunications (BUPT) under Grant 2024YQTD02; in part by Tianjin Natural Science Foundation General Project under Grant 23JCYBJC00780; in part by Tianjin Xinchuang Haihe Laboratory under Grant 22HHXCJC00002; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515140137; in part by the National Research Foundation, Singapore, and Infocomm Media Development Authority through Future Communications Research and Development Programme, Defence Science Organisation (DSO) National Laboratories under the artificial intelligence (AI) Singapore Programme under Grant FCP-NTU-RG-2022-010 and Grant FCP-ASTAR-TG-2022-003, Singapore Ministry of Education (MOE) Tier 1 under Grant RG87/22, the Nanyang Technological University (NTU) Centre for Computational Technologies in Finance (NTU-CCTF), and Seitee Pte Ltd; in part by the RIE2025 Industry Alignment Fund - Industry Collaboration under Project IAF-ICP and Project I2301E0026; in part by the Administered by A*STAR, as well as supported by Alibaba Group and NTU Singapore through Alibaba-NTU Global e-Sustainability CorpLab (ANGEL); and in part by the Ministry of Education, Singapore, through Academic Research Fund Tier 2 under Grant MOE-T2EP20221-0017, Singapore Management University (SMU)-Singapore University of Technology and Design (SUTD) Joint Grant 22-SIS-SMU-048, and SUTD Kickstarter Initiative under Grant SKI 20210204. *(Corresponding authors: Dusit Niyato; Xiaofei Wang.)*

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Digital Object Identifier 10.1109/JPROC.2024.3520803

NOMENCLATURE

AI	Artificial intelligence.
BSM	Bell state measurement.
DAOs	Decentralized autonomous organizations.
DAPPs	Decentralized applications.
DLT	Distributed ledger technology.
DID	Decentralized identifier.
EMRs	Electronic medical records.
IoT	Internet of Things.
IPFS	Interplanetary file system.
ML	Machine learning.
NFT	Nonfungible token.
PoA	Proof of authority.
PoE	Proof of entanglement.
PoS	Proof of stake.
PoW	Proof of work.
PQB	Postquantum blockchain.
PQC	Postquantum cryptography.
QDPoS	Quantum delegated proof of stake.
QKD	Quantum key distribution.
UAVs	Unmanned aerial vehicles.
UGC	User-generated content.

I. INTRODUCTION

A. Background

The decentralized digital economies in Web 3.0, including creator economies and DAOs, can improve the user experience and take user privacy and security to the next level [1]. Without trusted authorities, users own their unique digital identities built on blockchain to access personal data and DAPPs in Web 3.0. In addition to read-and-write operations, Web 3.0 can provide proof-of-ownership for digital assets of participants based on blockchain technologies. In Web 3.0, users' data are stored in or linked to the blockchain and stored in distributed data storage systems such as the IPFS [2]. Meanwhile, activities and regulations in Web 3.0 are executed via immutable and transparent smart contracts and consensus algorithms proposed by DAOs. Web 3.0's decentralized nature empowers users with greater control over their data, which reduces reliance on the management of centralized authorities [1]. The role of blockchain is to enhance the security and transparency of Web 3.0 by providing a trustworthy way for users to store data and conduct transactions without the need for centralized authorities. However, it is essential to note that blockchain technology is susceptible to quantum computing, which can break one-way mathematical functions such as hash functions and public-key encryption commonly used in classical computing [3]. Due to the superposition and entanglement of quantum bits (qubits), quantum computing has enough processing power to break existing digital signatures and hash functions within secrecy periods. For example, a quantum computer with 2×10^7 qubits can crack Rivest–Shamir–Adleman (RSA)-2048 in just eight hours [4]. Hence, it is possible for a malicious user utilizing quantum computers to easily steal private keys from a blockchain wallet, ultimately resulting in the loss of all funds in that wallet.

While Web 3.0 affords users the ability to peruse, produce, and possess their UGC, traditional blockchain technology is on the verge of obsolescence once the potency of a quantum computer meets the prerequisites to operate Shor's algorithm and Grover's algorithm [3]. The proliferation of quantum information technologies has opened a new avenue for blockchain technology in Web 3.0. A quantum blockchain, which is founded on quantum information technologies, can formulate more secure and resilient distributed ledgers that are impervious to quantum attacks. For instance, QKD protocols encode the key employed for symmetric encryption into a quantum state for transmission over a quantum channel. With one-time-pad technology, QKD-secured communication can realize information-theoretically secure data transmission. Furthermore, quantum blockchain can utilize the quantum hash function and one-way quantum computing functions to fabricate quantum voting [5] and quantum signature [6] algorithms. Additionally, PQB, which is based on postquantum cryptographic algorithms that are immune to quantum attacks, can also be deemed as a solution to secure Web 3.0 in the postquantum era. Consequently, classical blockchain is revolutionized into the quantum blockchain with the added benefits of prolonged data encryption and over two decades of usage.

Due to the substantial current interest and complementary advantages of quantum information technologies and blockchain, we aim to examine how state-of-the-art quantum information technologies are driving blockchain to open up new horizons for the development of Web 3.0 from the ambilateral perspective. Specifically, we explore the benefits of quantum blockchain in terms of internal technologies, i.e., consensus protocols, incentive mechanisms, smart contracts, and cryptographs. In addition, we also investigate how quantum technologies can enhance the performance of blockchain in terms of external aspects such as decentralization, profitability, and privacy of storage.

B. Related Books and Surveys

Recently, there have been many efforts that attempt to converge quantum information technologies and blockchain into the quantum blockchain and PQB. Meanwhile, the growth of quantum information theory and computation has resulted in a rise in the number of research ongoing in the quantum blockchain.

Numerous researchers have examined the integrated frameworks, potential opportunities, and inherent challenges within the field of quantum blockchain [7], [8], [9], [10], [11], [12], [13], [14]. Faridi et al. [7] and Krishnaswamy et al. [8] investigate the possibilities and potential benefits of quantum computing in blockchain systems. The advantages of quantum blockchain and development prospects are summarized in [9] compared with

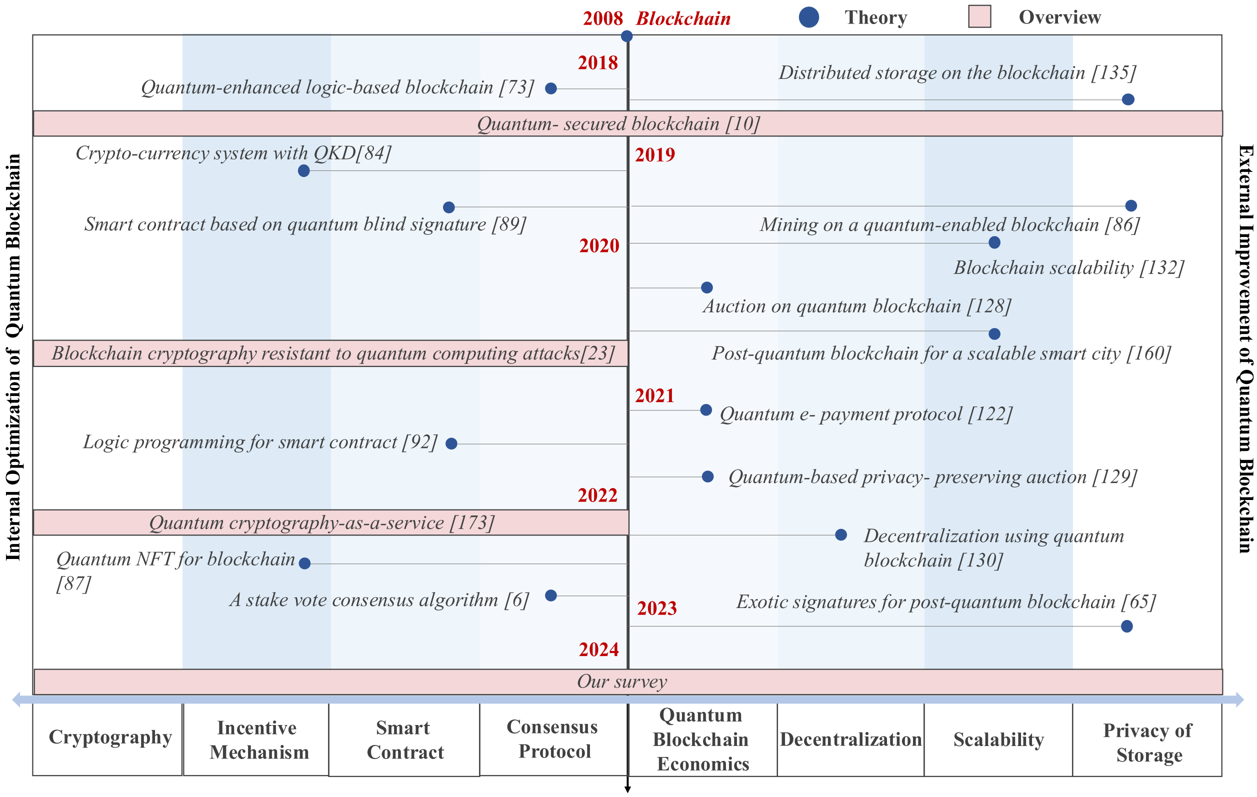

Fig. 1. ***Research activities of quantum-driven blockchain in Web 3.0.***

the classical blockchain while presenting the detailed method of applying quantum technology to the blockchain. Kiktenko et al. [10] provide an overview of the current state of research on quantum blockchain, covering topics such as quantum-resistant blockchain protocols and cryptography algorithms. Rajan and Visser [11] explore the potential of using entanglement in time to create a quantum blockchain and present the challenges of implementing a quantum blockchain. Further, Dey et al. [12] provide an overview of the challenges and opportunities associated with developing a quantum blockchain, including issues related to quantum security, scalability, and interoperability. Allende et al. [13] propose some tutorials on the implementation of quantum blockchain. Gharavi et al. [14] investigate the potential utilization of PQBs in designing security measures for diverse IoT applications.

In this article, we summarize research studies of quantum-driven blockchain in Web 3.0 as referenced in [15] and illustrated in Fig. 1. We primarily discuss the role of quantum blockchain in supporting Web 3.0 through internal technology optimization and external performance improvement. Specifically, internal optimization enhances core blockchain elements, i.e., consensus protocols, cryptography, smart contracts, and incentives with quantum technologies to enhance security, efficiency, and scalability against quantum threats. External improvements focus on how quantum technologies enhance blockchain's interaction with its environment, advancing decentralization, profitability, and privacy. We list details of the existing research works and our survey in Table 1.

Unless quantum technology is integrated into blockchain technology, the components of classical blockchain are exposed to quantum attacks [3], resulting in security risks to the internal technologies and external performance of the blockchain. Currently, we have already seen the development of information-theoretically secure quantum protocols, such as quantum homomorphic encryption protocols [16], [17], quantum image encryption [18], [19], and quantum digital signature [20], [21], while several attempts have been made to incorporate PQC [22], [23]. These protocols bring benefits to the internal technologies and external performance of blockchain, achieving privacy, security, effectiveness, and profitability under quantum attacks.

Nevertheless, the above studies focus on either internal technical optimization or external performance improvement of quantum technologies for blockchain in Web 3.0. There is no research that explores the benefits of quantum technology for blockchain in both aspects. Therefore, this article provides a comprehensive account of blockchain combined with quantum information technology from both internal and external aspects in a Web 3.0 scenario, starting from an overview, motivation, and comprehensive framework to research challenges and future directions.

Table 1 Details of the Existing Surveys

Ref.	Quantum Technology	Benefits of Blockchain	Topic Covered	Consensus Mechanism	Smart Contract	Decentralization	Privacy
[7]	Quantum mechanics	Internal technologies	Present some potential benefits of quantum computing in blockchain systems	✓	✓	×	×
[8]	Quantum communication	Internal technologies	Explore the possibilities of quantum blockchain systems	×	✓	×	×
[9]	Quantum computing	Internal technologies	Analyze the advantages and development prospects of quantum blockchain	✓	×	×	×
[10]	Quantum key distribution	External performance	Propose a possible solution to the quantum blockchain challenge	✓	×	×	×
[11]	Quantum entanglement	External performance	Propose a conceptual design for the quantum blockchain	×	×	×	✓
[12]	Quantum cryptography	External performance	Discuss challenges and opportunities to develop a quantum blockchain	×	×	×	✓
[13]	Quantum-safe cryptography	Internal technologies	Give some implementation tutorial of quantum blockchain	×	×	×	✓
[14]	Post-quantum cryptosystem	Internal technologies	Exploring the evolution of post-quantum blockchains and their potential utilization in security frameworks	×	×	✓	✓
Our survey	Quantum information technologies	Internal technologies/ External performance	Provide a comprehensive account of blockchain in combination with quantum information technology in Web 3.0	✓	✓	✓	✓

C. Our Survey

Advances in quantum computers undermine the security of classical blockchains; there is an urgent need to solve the limitations of blockchain and continue to develop Web 3.0. This article aims to integrate quantum information technologies into blockchain and investigate the most recent developments in Web 3.0 thoroughly. The contributions of this article can be summarized as follows.

1) First, we present the basic principles of blockchain and quantum information technology while giving motivations for using the complementary properties of quantum technology to support blockchain from the ambilateral perspective, i.e., internal technologies optimization and external performance improvement.
2) Second, we optimize the internal technologies of quantum blockchain from four aspects, including consensus protocols, incentive mechanisms, smart contracts, and cryptography.
3) Third, we improve the external performance to better support quantum blockchain, including economy, decentralization, scalability, and privacy of storage.
4) Fourth, we investigate potential quantum blockchain applications and give tutorials for implementing quantum blockchain in Web 3.0.
5) Finally, we explore several key challenges and open research directions.

A taxonomy graph of this article is presented in Fig. 2. Specifically, we first give the background on blockchain and quantum information technologies in Sections II and III. Then, the motivation for integrating them is explained in Section IV. Next, we describe, in detail, the integration techniques from internal technologies optimization (Section V) and external performance improvement (Section VI). In Section VII, we investigate some applications and tutorials of quantum blockchain. Section VIII discusses research challenges and future directions. Lastly, the conclusion is drawn in Section IX.

II. OVERVIEW OF BLOCKCHAIN IN WEB 3.0

In this section, we present the background of blockchain in Web 3.0, including the definition, internal technologies, and external performance.

A. Fundamental and Enabling Infrastructure of Web 3.0

As the cornerstone of the digital economy, Web 3.0 drives digital transformation in a decentralized, secure, and transparent way. The read/write/own Web 3.0 has evolved from the read-only Web 1.0 and the interactive Web 2.0, allowing content creators and users to own and manage their digital assets [1]. In Web 1.0, users could only read content published by authoritative sources, with data and value concentrated in the hands of Internet owners, stored on local servers. Web 2.0 introduced platforms and applications that let users access, create, and share content, but data ownership remained with companies on cloud servers, limiting users' control over their own content. In Web 3.0, users access the Internet through digital identities and wallets, enabling them to manage and own their data in online services. Dubbed the Internet of Value [24], Web 3.0 allows users to create, manage, and trade UGC as digital assets. Supporting this evolution requires a decentralized, secure, and transparent infrastructure for data, computing, and networking, essential for establishing consensus and running DAPPs with smart contracts.

Blockchain, based on consensus algorithms and smart contracts, enables decentralization and interoperability for

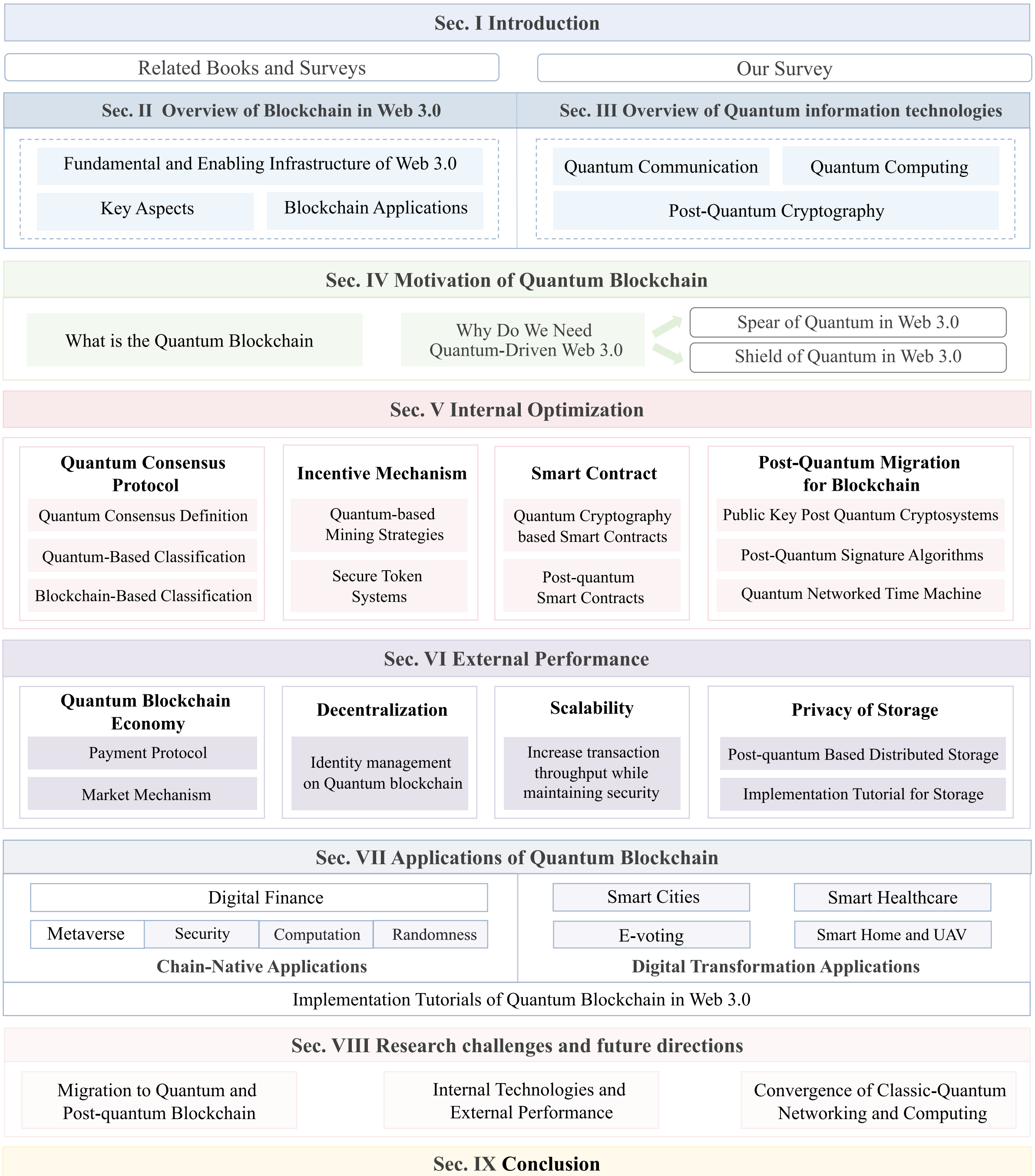

Fig. 2. *Taxonomy graph of quantum-driven blockchain in Web 3.0.*

Web 3.0 applications [25]. In Web 3.0, each user has a digital wallet holding their digital identity, allowing them to interact with distributed ledgers to manage data assets, digital assets, and transaction histories, all recorded anonymously on-chain. Distributed storage supports Web 3.0 by storing user interaction and authentication data across multiple nodes, reducing storage costs [2], enhancing security, and preventing data loss through replication. This decentralized approach is more secure and resilient than traditional centralized storage.

In addition to distributed data storage, heterogeneous computing nodes, such as cloud data centers, edge servers, and endpoints, provide computing power for Web 3.0 applications [26]. In Web 3.0, participants are both users and creators. To maintain decentralization in the digital society, on the one hand, users need to contribute their computing resources to reach a consensus and execute smart contracts. On the other hand, they must use computational resources to run creative applications to create their new content, such as AI models to synthesize images.

In addition, to store the created content as digital assets, computational power is also needed to store created content as digital assets and to enable smart contract-based voting within DAOs [27]. Based on DAOs, community activities in Web 3.0, such as task assignment, payroll, and organizational governance, are open, transparent, and autonomously managed through blockchain.

B. Key Aspects of Blockchain in Web 3.0

As mentioned above, blockchain-based digital ecosystems with distributed data storage and computing-power networks are the foundation for the digital society of Web 3.0. Notably, Web 3.0 is considered to be a collection of blockchain-based protocols focused on changing the ecology of the Internet. This article describes the key aspects of blockchain in Web 3.0 in terms of internal technologies and external performance.

1) Internal Technologies: It refers to the support of blockchain for Web 3.0 in terms of internal composition.

1) *Consensus protocol:* Consensus protocols are essential to decentralized Web 3.0, enabling agreement among multiple parties without third-party authorities, and ensuring network security and reliability. Common protocols include PoW [28], PoS [29], and PoA [30], each suited to different Web 3.0 applications. For example, DAOs select suitable protocols according to the various requirements of users and applications.
2) *Incentive mechanism:* As the Internet of Value, Web 3.0 offers various incentives to encourage participation, including miner rewards, transaction fees, and minting fees, distributed based on blockchain platforms. In addition, blockchain incentives allow participants to create and manage digital assets as NFTs [31], enabling the tokenization of digital identities, intellectual property, and real-world assets. These incentives are essential to Web 3.0's economic design, motivating fair and transparent participation in the ecosystem [32].
3) *Smart contract:* As the foundation of Web 3.0, smart contracts are self-executing code that powers a range of functions, from financial transactions and user identification to running DAPPs [33]. These contracts set the rules for Web 3.0 interactions, allowing users to engage with DAPPs on the blockchain and enabling decentralized transactions without third-party verification. Use cases include decentralized finance (DeFi), digital identity, and gaming, where smart contracts combine tokens or crypto rewards with gameplay in GameFi and Play-to-Earn models, ushering in a new era of entertainment and economy in Web 3.0.
4) *Cryptography:* Cryptography is a method of storing and transmitting data in a secure form to prevent third parties from accessing private messages during peer-to-peer communication. In blockchain, this aspect is essential to Web 3.0, protecting user data and privacy [34]. Furthermore, cryptography ensures the security of participants and transactions, preventing double-spending and maintaining transaction integrity in Web 3.0.

2) External Performance: It refers to the support of blockchain for Web 3.0 in terms of external properties.

1) *Decentralization:* As mentioned above, one of the core issues with Web 2.0 is the centralized management of authority and data. Blockchain addresses this limitation by decentralizing Web 2.0 into Web 3.0, enabling broader distribution of both data and authority. Specifically, Web 3.0, also known as the decentralized network, is a new version of the Internet based on blockchain technology. Blockchain serves as a decentralized digital ledger that uses encryption to safeguard and verify transactions. By adopting a blockchain-driven public distributed ledger, Web 3.0 enhances transparency and decentralization, fostering a more open, transparent, and secure Internet [35].
2) *Profitability:* One of the most significant properties of Web 3.0 is profitability. Web 3.0 disrupts the DAPPs industry by tokenizing assets. This process involves the creation of digital tokens, which can be bought, sold, and traded on blockchain-based platforms, allowing investors to diversify their portfolios and increase their potential profits. DeFi is an emerging financial system that provides financial services and products in a blockchain network, providing investors with alternative financing options for transactions [15]. Nexo is one such platform for real estate transactions [36], enabling real estate investors to use their property as collateral for loans, thus offering greater flexibility, liquidity, and the potential for higher profits.
3) *Privacy and security:* Blockchain plays a crucial role in ensuring the privacy and security of data storage and identity management. From Web 1.0 to Web 2.0, significant progress has been made in decentralizing data, empowering users to create and share data, whereas users do not own and control their data [37]. Further, Web 3.0 provides users with an open, credible, and permissionless Internet to ensure data ownership and privacy. Web 3.0 enhances the privacy and integrity of user data, and due to the immutability of blockchain, data within this framework remain transparent and permanent, supporting secure identity management.

Therefore, it is quite clear that blockchain can serve as a key driver of the next generation of the Internet. In the following, we present some blockchain applications in Web 3.0.

C. Blockchain Applications in Web 3.0

1) DAPPs: DAPPs run on blockchain networks and smart contracts instead of traditional centralized servers [38].

Furthermore, DAPPs allow users to interact in a more secure and private way because the transaction records on the blockchain network are public, tamper-proof, and traceable. Some researchers demonstrate that DAPPs in Web 3.0 can replace the existing centralized Web 2.0 platforms by allowing users to use blockchain-based digital identities across different DAPPs. Meanwhile, DAPP developers often incentivize user participation through financial rewards, such as digital currencies or NFTs. DAPPs have diverse applications, including digital currencies, gaming, financial services, and social networking. To enable interactions and offer incentives, many DAPPs rely on tokens, typically cryptocurrencies.

2) DAO: DAO stands for an organization that is governed by rules encoded as computer programs called smart contracts and operates on a decentralized blockchain network, which provides a high degree of security, transparency, and immutability [39]. DAO aims to enable decentralized decision-making and management, allowing individuals to collaborate and coordinate. The decision-making process in a DAO is usually carried out through a voting mechanism, where token holders can vote on proposals and decisions that affect the organization's future. Web 3.0 considers that users and online communities can interact and organize through DAO. DAO has been used in a variety of scenarios, including crowdfunding, investment, and community management. Overall, DAO is still in the early stages of development, and it has the potential to disrupt traditional organizational structures and enable new forms of decentralized decision-making and governance.

3) NFT: Based on blockchain technology, NFT represents a unique digital asset rather than a fungible currency [32]. Specifically, NFT leverages blockchain technology to create, sell, and trade digital assets due to its decentralized, transparent, and tamper-evident properties. In Web 3.0, NFT and digital currencies can be purchased online, represent digital ownership, make payments, and access specific services and applications. Tokens of this type can be bound to virtual/digital properties as their unique identification. With NFT, all marked properties can be freely tradable with customized values based on their ages, liquidity, etc. It has greatly stimulated the prosperity of digital art markets and is increasingly popular in collections, gaming props, and physical asset traceability.

III. OVERVIEW OF QUANTUM INFORMATION TECHNOLOGIES

In this section, we give the background of quantum information technologies, including quantum communication, quantum computing, and PQC, as shown in Fig. 3.

A. Quantum Communication

In the Internet of Everything era, huge amounts of sensitive data rely on communication networks. These data

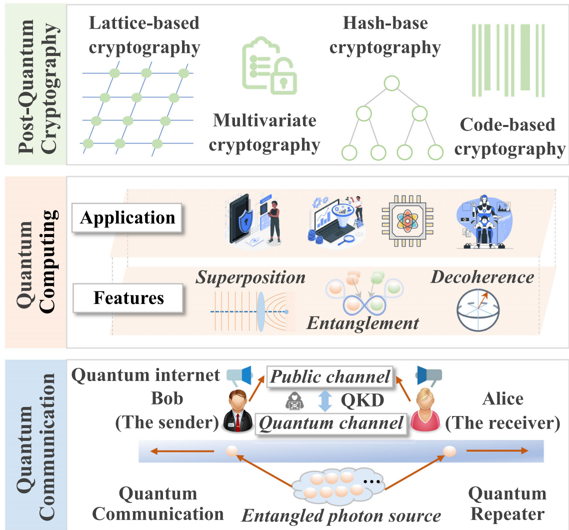

Fig. 3. ***Quantum information technologies.***

are usually encrypted and then sent along with a "key" over fiber-optic cables. In particular, these data and keys are sent as classical bits, which can be read and copied by hackers without leaving a trace, thus causing security problems. In the following, we give some key terms to fully understand quantum communication.

1) *Quantum:* Quantum in "quantum computing" means the quantum mechanics used by the system to calculate the output. Generally, a quantum refers to the smallest possible discrete unit of any physical property, e.g., electrons, neutrinos, and photons.
2) *Qubit:* Similar to the bits in classical computing, qubits are the basic unit of information in quantum computing. The difference between them is that a classical bit is binary and it can only take the value of 1 or 0, while a qbit can maintain a superposition of all possible states.

Quantum communication leverages the laws of quantum physics to protect private data [40]. These principles enable particles, such as photons used for data transmission, to exist in a superposition state. That means the measurement result of such states is not definitively $|0\rangle$ or $|1\rangle$. Here, $|\cdot\rangle$ is called a "ket-vector" in Dirac notation, which is used to denote a vector in a complex vector space (the Hilbert space) that describes the state of a quantum system. It also means particles can take on multiple combinations of values of 1 and 0 simultaneously. These particles are called qubits, or qbits. The super-fragile quantum state of qubits causes hackers to be unable to tamper with them without leaving signs of activity. Quantum communication focuses on the generation and use of quantum states for communication protocols. Significant progress has been achieved, with researchers actively developing quantum-secure encryption protocols and harnessing quantum properties to establish highly secure communication channels and global quantum networks. Current efforts are focused on the following key directions.

1) Quantum Teleportation: Quantum teleportation is a process by which a qubit can be transmitted from a sender to a receiver, without that qubit actually being transmitted through space [41]. Unlike the portrayed teleportation in science fiction, which is described as a means of transferring physical objects from one location to another, quantum teleportation only transmits quantum information [42].

Without loss of generality, we analyze the process of quantum teleportation between Alice and Bob in the following. The data qubit state from source node Alice $|\phi_D\rangle = \alpha|0\rangle + \beta|1\rangle$, as the input, is to be transmitted from Alice to Bob, where the coefficients of $|0\rangle$ and $|1\rangle$ represent the probability amplitudes. Einstein, Podolsky, and Rosen (EPR) pairs of Bell state $|\Phi^+\rangle$, raised by three scientists Einstein, Podolsky, and Rosen, are introduced to be in possession of both Alice and Bob. The Bell pair qubit held by Alice is $|\Phi^+\rangle_A$, while the other pair of qubits that Bob holds is denoted by $|\Phi^+\rangle_B$. The detailed process of teleportation between Alice and Bob is given as follows.

1) EPR pairs of Bell state $|\Phi^+\rangle$ are introduced for the teleportation of Alice and Bob such that Alice holds $|\Phi^+\rangle_A$ and Bob holds $|\Phi^+\rangle_B$ state.
2) Alice obtains the state to be transmitted, i.e., $|\phi_D\rangle$.
3) Bell state and data qubit as a measurement $|\phi_D\rangle|\Phi^+\rangle_A$ are performed by Alice. The measured result is stored in the two classical bits and then sent to Bob through the link.
4) Then, Bob operates on the received classical bits.
5) By now, Bob's state of qbit $|\Phi^+\rangle_B$ has the same quantum state with Alice $|\phi_D\rangle$. The qubits transmission between Alice and Bob's end nodes is completed.

Moreover, quantum teleportation uses quantum entanglement to transfer a particle's state to another particle [43]. This process, involving entanglement generation and distribution through quantum and classical channels, has been demonstrated with photons, spins, and superconducting qubits, enabling quantum information exchange between qubits.

2) Quantum Key Distribution: As the central focus in quantum communication, QKD uses quantum teleportation to create key pairs for encrypting and decrypting messages [44]. Various approaches and protocols now exist for implementing QKD, which prevents hacking, protects privacy, and addresses the key distribution problem through quantum cryptography [45]. Here, we use the BB84 protocol as an example to illustrate the QKD process, assuming Alice wants to securely send data to Bob.

1) *Generation and transmission of encryption key:* Alice creates an encryption key in the form of qubits. Then, these qubits would be sent to Bob by the fiber-optic cable.
2) *State measurements of qbits:* Bob then uses a random base to measure the qbits received with the basis, after which Bob acknowledges to Alice the qubits he received through the public channel.
3) *Key sifting:* Alice and Bob can determine that they hold the same key by comparing the state measurements of a fraction of the qbits.
4) *Key distillation:* It is run by Alice and Bob, involving calculating whether the error rate is high enough to indicate that the hacker is trying to intercept the key.
5) *Key update:* If so, the suspect key might be threw away while the new keys are generated until Alice and Bob are confident that they share a secure key. Then, Alice can encrypt the data with her key and send it as classical bits to Bob, who uses his key to decode the message.

The first QKD experiment in 1989 achieved secure transmission over 32 cm [46]. To enable QKD over long distances using fiber optics or free-space channels, this range must be extended. However, longer distances increase losses due to decoherence, reducing photon count and key rates. Currently, the longest QKD network spans 2032 km between Beijing and Shanghai in China, used by banks and financial institutions for secure data transmission [47]. QKD would be even more secure with the addition of quantum repeaters.

3) Quantum Repeater: Due to signal loss and decoherence, photons in quantum communications typically travel only a few tens of kilometers. Allocating quantum resources, such as entanglement and qubits, over long fiber optic networks is challenging since quantum signals cannot be amplified as in classical networks. Quantum repeaters, designed to overcome transmission loss, help maintain encryption keys in quantum form over long distances but require complex devices [48]. Recent advances have integrated quantum repeaters into standard fiber networks, improving secure quantum communications and supporting QKD advancements.

Similar to classical repeaters, quantum repeaters divide the transmission distance into segments [49], with entangled photons sent to quantum memory and BSM devices. This allows entanglement to be shared between distant parties, extending the communication range. Quantum memory stores photons until the next entanglement link is ready, ensuring they remain unmeasured. Re-emitting photons between quantum memories for BSM further extends distances. Quantum memory remains a major research focus to improve storage time and retrieval efficiency [50]. However, integrating these fundamental elements into a unified quantum communication network is a key challenge moving forward.

4) Quantum Internet: The quantum Internet consists of quantum computers that send, compute, and receive information encoded in quantum states [51]. Similar to the traditional Internet, it would span the globe, but with a quantum-based communication network. Rather than replacing the traditional Internet, the quantum Internet would offer new functions, such as quantum cryptography and quantum cloud computing [52]. Quantum computers communicate through quantum states, enabling them to solve problems such as factoring large numbers or complex logistics, especially useful for cryptography. Connected via the quantum Internet, these computers could combine their power for enhanced performance.

Significant progress has been made toward a quantum Internet. In 2017, China launched the Micius satellite, which enabled the first QKD-based intercontinental video conference between Beijing and Vienna [53]. China is also planning additional quantum satellites to expand its QKD network. In 2019, researchers in the United States established a 10-mil quantum network, which has since expanded to an 80-mil testbed [54]. In 2022, Chicago added a 35-mil extension, marking progress toward an interstate quantum network [55]. The quantum Internet will not replace the traditional Internet; instead, it will support organizations needing secure data handling, while most regular information, such as photographs and videos, will still be shared via classical bits. However, the quantum Internet may face quantum-based attacks, making the development of secure quantum networking solutions essential.

B. Quantum Computing

1) Definition of Quantum Computing: Classical computers encode information into bits using a binary stream of electrical pulses (1 and 0), limiting their processing ability. In contrast, quantum computers use qubits, which can represent 1, 0, or a complex combination of both 1 and 0, enabling complex calculations beyond classical capabilities. This ability to observe quantum systems and solve intricate probabilities accelerates advancements in AI and Web 3.0. Quantum computing leverages quantum mechanics to solve problems that are too large or complex for classical computers [56]. By leveraging quantum mechanics with phenomena such as superposition, entanglement, and interference, quantum computing introduces new programming capabilities.

2) Features of Quantum Computing: Quantum computing possesses distinctive properties that enable not only faster and more efficient processing but also the exploration of previously unimaginable computational possibilities.

1) *Superposition:* Quantum particles exist in a superposition, meaning each qubit can represent both 0 and 1 simultaneously [57]. Unlike classical bits, which yield a fixed outcome such as heads or tails on a coin, a qubit can show both states at once. Groups of qubits in superposition can tackle complex, multidimensional problems.
2) *Entanglement:* Entanglement allows qubits, even if separated by vast distances, to instantly influence each other [58]. When two qubits are entangled, altering one impacts the other, regardless of the distance between them, as long as the particles remain isolated.
3) *Decoherence:* Decoherence occurs when qubits interact with the environment, leading to photon loss and errors in computing [48]. This causes qubits to lose their superposition state, reducing quantum fidelity, which is a key measure of reliability in quantum systems.
4) *Superconductors:* At ultralow temperatures, certain materials exhibit quantum effects similar to superconductivity, where electrons move without resistance [59]. In superconductors, electrons pair up to form "Cooper pairs" and, placed across a barrier, create a Josephson junction.
5) *Control:* Quantum computers use Josephson junctions as superconducting qubits [60]. By directing microwave photons at these qubits, their states can be held, changed, and read, allowing the manipulation of individual quantum information units.

3) Applications of Quantum Computing: Quantum computing has been applied in numerous applications due to its advantages and properties.

1) *Quantum simulation:* Quantum computers leverage quantum properties in their calculations and thus perform exceptionally well in simulating other quantum systems [61], including photosynthesis and superconductivity. This capability allows them to tackle problems that are too complex or ambiguous for classical computers to handle effectively.
2) *Cryptography:* Classical cryptography, such as the RSA algorithm [62], depends on the intractability of problems (i.e., integer factorization or discrete logarithms). Many of these problems can be addressed more efficiently by quantum computing.
3) *Optimization:* Some complex logistics calculations can be solved by running quantum-based optimization algorithms on a classical computer, which, in turn, can find solutions that were previously impossible. Quantum computing helps us find better ways to manage complex systems such as traffic flow, traveling salesman problem and package delivery [63].
4) *Quantum ML:* Quantum ML aims to devise and implement quantum software that enables faster ML [64]. In quantum ML, the qubits and quantum operations or specialized quantum systems are leveraged to improve computational speed and data storage accomplished through algorithms in a program. Some novelty quantum algorithms are proposed that could used as the building blocks of ML programs but also place greater demands on hardware and software.

C. Postquantum Cryptography

Quantum computers pose a serious security threat to many commonly used symmetric encryption algorithms and key negotiation schemes due to their ability to exponentially increase computational power [65], [66]. This heightened computational power significantly raises the risk of breaking current network security encryption, prompting governments and companies to shift toward postquantum encryption methods. After years of research, PQC, including lattice-based, multivariate, hash-based, and code-based techniques, has emerged as a highly developed and reliable approach grounded in classical computing.

1) *Lattice-based cryptography:* It utilizes the mathematical structure of lattices in an n-dimensional space to provide secure cryptographic solutions [67]. In blockchain, it can create digital signature schemes to authenticate and verify transactions, offering resistance to both classical and quantum attacks. Additionally, it supports privacy-preserving protocols, such as zero-knowledge proofs, enabling privacy-focused applications such as confidential smart contracts and private transactions.
2) *Multivariate cryptography:* It utilizes multivariate polynomial equations to create cryptographic systems such as digital signatures and public-key encryption [68]. In these systems, the secret key consists of polynomial coefficients, and security relies on the difficulty of solving multivariate equations. While it could secure blockchain transactions, multivariate cryptography is vulnerable to quantum attacks and is not considered a robust postquantum solution, with lattice-based and code-based cryptography seen as more secure for blockchain.
3) *Hash-based cryptography:* It relies on cryptographic hash functions to generate cryptographic primitives such as digital signature schemes and key derivation functions [69]. Its security relies on the collision resistance of the hash function, making it hard to find two inputs with the same hash output. In blockchain, hash-based cryptography secures transactions and data through hash-based signatures and key derivation methods.
4) *Code-based cryptography:* It utilizes error-correcting codes to generate cryptographic keys to secure communication [70]. These keys have multiple applications, including securing communication between parties in a blockchain system. Code-based cryptography can protect communication between network nodes, authenticate exchanged messages, and establish secure connections. Additionally, code-based keys can secure blockchain data by creating digital signatures for transaction verification.

Quantum blockchain research in Web 3.0 aims to ensure a smooth transition from classical to PQC to prevent shutdowns and economic loss if quantum computers eventually break traditional cryptographic systems. In this article, we provide a summary of the key definitions, with the corresponding abbreviations presented in the Nomenclature.

IV. MOTIVATION OF QUANTUM BLOCKCHAIN IN WEB 3.0

To date, the limitations of blockchain and the complementary advantages of quantum information technology are obvious. Naturally, the emergence of quantum information technology-assisted blockchain is expected to have a significant impact on Web 3.0. In this section, we mainly discuss the motivation of quantum blockchain in Web 3.0, as shown in Fig. 4.

A. What Is the Quantum Blockchain

Currently, there is no standard definition for quantum blockchain. Shrivas et al. [71] consider quantum blockchain as a distributed, decentralized, and cryptographic database based on quantum information theory and computation. Nilesh and Panigrahi [72] demonstrate that quantum blockchains should satisfy the following properties: 1) decentralized architecture; 2) a quantum network with a distributed ledger; 3) nodes in a quantum network need to possess quantum capabilities, e.g., quantum storage and quantum state preparation; and 4) shared quantum database. Compared with classical blockchains, quantum blockchains have the following advantages [73]: 1) more efficient; 2) more powerful; 3) more secure; 4) cheaper; 5) smarter; and 6) easier to regulate.

In this article, we demonstrate that quantum blockchain in Web 3.0 is a technology that combines the principles of quantum information technologies and blockchain, aiming to create a more secure and efficient ecosystem for Web 3.0. Actually, we categorize the quantum blockchain architecture into three distinct layers, as shown in Fig. 4.

1) *Infrastructure layer:* It integrates blockchain and quantum technologies, providing essential support. This includes managing the distributed ledger and network protocols for blockchain while using quantum cryptography, QKD, and quantum-enhanced processing to improve security, efficiency, and scalability.
2) *Components layer:* It optimizes internal technologies and enhances blockchain external performance. This layer leverages technologies from the infrastructure layer to ensure the blockchain operates efficiently, securely, and reliably while addressing real-world needs.
3) *Application layer:* This layer enables various applications, including smart homes and smart healthcare, by leveraging integrated technologies designed for a Web 3.0 environment. It ensures seamless and efficient user interaction with the quantum blockchain, delivering secure, scalable, and user-friendly functionalities.

B. Why Do We Need Quantum Blockchain in Web 3.0

1) Limitation of Blokchain: The motivation for quantum blockchains in Web 3.0 stems from the limitations

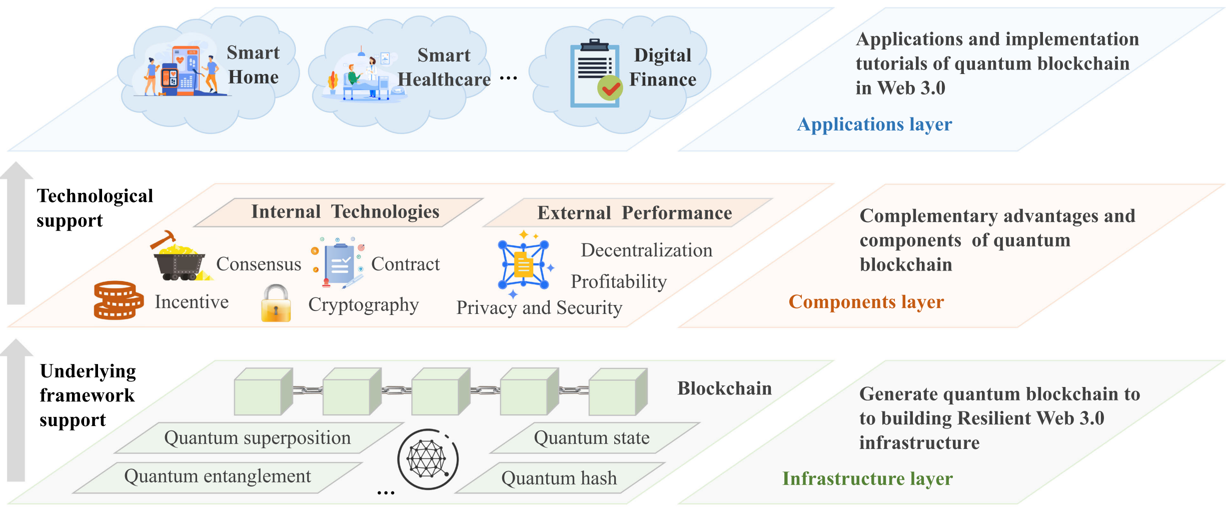

Fig. 4. ***Motivation of quantum-driven blockchain in Web 3.0.***

of traditional blockchains, which may be vulnerable to attacks from quantum computers. Next, we describe the limitations of blockchain in two aspects.

a) Internal technologies: The internal compositions of blockchain face many challenges.

1) *Consensus protocol:* The most prominent shortcoming of consensus protocol is the high consumption of computational resources, resulting in low scalability and long latency for Web 3.0 services. Moreover, some protocols may suffer quantum attacks. Users with quantum computers can calculate hash values, making 51% of attacks possible.
2) *Incentive mechanism:* There are certain elementary risks associated with distributed digital currencies, such as double-spending, sybil attacks, and eclipse attacks.
3) *Smart contract:* The powerful quantum computation poses serious security threats to the smart contract. Meanwhile, creating a programming language for smart contracts in quantum logic deserves close attention and serious consideration.
4) *Cryptography:* With the advent of quantum computers, the main issue with classical blockchain is the possibility of cracking hash functions and classical digital signatures based on asymmetric cryptography.

b) External performance: The external properties face many challenges.

1) *Decentralization:* The decentralized architecture is subject to security threats, such as malicious attacks and transaction tampering. Although blockchain technology can provide anonymity, it may also raise privacy concerns.
2) *Profitability:* Classical cryptography in payment and voting protocols is threatened by quantum computers, raising economic issues of anonymity, verifiability, fairness, profitability, etc., in Web 3.0 scenarios.
3) *Privacy and security:* Since blockchain nodes store a copy of all blocks, storage and the privacy of stored data are huge problems as the number of blocks in the blockchain continues to grow.

2) Benefits of Quantum Information Technologies: Despite the limitations of blockchain, quantum blockchain is also motivated by the benefits of quantum information technologies. Next, we describe these benefits in two aspects.

a) Internal technologies: Quantum information technologies bring benefits for internal compositions of blockchain.

1) *Consensus protocol:* The quantum consensus is more secure, efficient, and fair than traditional consensus protocols, avoiding high consumption of computational resources while providing higher throughput.
2) *Incentive mechanism:* Quantum blockchain provides effective mining strategies and a secure token system for incentives.
3) *Smart contract:* On the one hand, quantum computing and quantum communication ensure the security of smart contracts. On the other hand, smart contracts based on PQC prevent quantum attacks.
4) *Cryptography:* Some postquantum migration methods and antiquantum signatures offer potential ways to achieve strong and unforgeable security under quantum attacks.

b) External performance: Quantum information technologies bring benefits to external properties of the blockchain.

1) *Decentralization:* Decentralized quantum blockchain tackles the issues caused by centralized management, providing a solution for achieving decentralization.
2) *Profitability:* Combining classical cryptography and quantum theory, quantum cryptography has emerged to ensure the absolute security of payment protocols.

Quantum technology enables anonymous, verifiable, and fair voting protocols, promoting economic prosperity and increasing profits in quantum markets.

3) *Privacy and security:* Some quantum/postquantum encryption and verification algorithms improve the communication costs, robustness, and privacy of distributed storage.

Due to the complementary advantages brought by quantum information technologies, it is obvious that the converged quantum blockchain technology shows a path for implementing distributed Web 3.0 infrastructures. In the following sections, we present deployment tutorials of quantum blockchain in Web 3.0 from two aspects, including internal technologies optimization and external performance improvement.

V. INTERNAL TECHNOLOGIES OPTIMIZATION IN QUANTUM BLOCKCHAIN

A. Quantum Consensus Protocol

In this article, we discuss the classification of quantum consensus protocol for Web 3.0 in detail to better understand quantum consensus. In this article, the quantum consensus protocol refers to the improved consensus protocol in blockchain with the support of quantum technology.

1) Quantum-Based Classification: The quantum consensus protocols can be divided into the following two categories based on the quantum mechanical features.

a) Measurement-based consensus: According to the assumptions of quantum mechanics, the operation of measuring the quantum state causes the collapse of the quantum state. As yet, many theoretical and experimental results have shown that measurement-based quantum control is an effective way to manipulate quantum states. Under this premise, various consensuses for the quantum network are proposed.

With reference to the traditional consensus mechanism of blockchain and some properties of quantum mechanics (i.e., randomness and irreversibility), a novel consensus mechanism of quantum blockchain is proposed [74]. This consensus integrates the zero-knowledge proof and quantum measurement. Here, we present an example to demonstrate the steps of the quantum zero-knowledge proof protocol. David and Ben have a highly confidential secret number A. Without loss of generality, assume David is the side who is trying to prove that he has A to Ben. The steps of the protocol are as follows.

1) *Distribution of shared key:* The QKD protocol, such as the BB84 protocol, is used to generate a shared key for David and Ben.
2) *Quantum state preparation:* David prepares several pairs of EPR entanglement photons and sends another photon from each pair to Ben.
3) *Security check:* The measurement results between two parties help to check the security of the quantum channel.
4) *Measurement and result encoding:* The two parties measure and encode their results, respectively.
5) *Transmission of proof information:* David encrypts the classical bit string from encoding while sending the encryption result to Ben.
6) *Verification of proof information:* By comparing the classical bit strings generated by two parties, the proof information would be verified.
7) *Reciprocal verification:* By exchanging the roles of David and Ben, the protocol enables reciprocal verification.

Compared with the traditional consensus, the measurement-based consensus is more secure and avoids high consumption of computing resources while providing higher throughput.

b) QKD-based consensus: Recently, QKD was proven to be unconditionally secure and has been used in quantum blockchain schemes. In the quantum blockchain, each pair of nodes is connected by a classical channel and a quantum channel, further forming a QKD network. Then, each pair of nodes uses the QKD network to establish a private key sequence for secure communication.

As the quantum public keys distributed by a node to another are the same, they cannot deny their actions. However, Kiktenko et al. [10] leverage QKD to distribute keys for each pair of nodes for verification. In this way, the two-way communication requires $O(n^2)$ implementations to distribution $O(n^2)$ keys. In particular, these works cannot provide nonrepudiation because each pair of nodes shares the same key.

Based on the unconditional security provided by QKD, a quantum communication protocol is designed [75]. By quantum-secured communication, a set of semihonest participants distribute the sequences of correlated numbers. These participants then exchange some information to reach a consensus. In the further development of this protocol, low-dimensional entanglement can be considered to replace the key distribution.

2) Blockchain-Based Classification: Referring to [76], consensus protocols in blockchain are classified into two groups in Web 3.0. The first group is the proof-based consensus protocol that uses computational proofs to determine the winner-generating block, such as PoW. The second group is the vote-based consensus protocol. In this regard, the nodes in the blockchain have equal votes and reach a consensus, such as Practical Byzantine Fault Tolerance (PBFT). According to the blockchain-based classification, we classify the quantum consensus protocol in the same way.

a) Proof-based consensus: Traditional proof-based consensus in Web 3.0 is computationally intensive and time-consuming, making blockchain systems inefficient.

To improve the blockchain consensus, the postquantum threshold signature scheme is exploited [77], [78]. This scheme is to solve quadratic equations in a finite field, which is resistant to attacks by quantum computers. Based

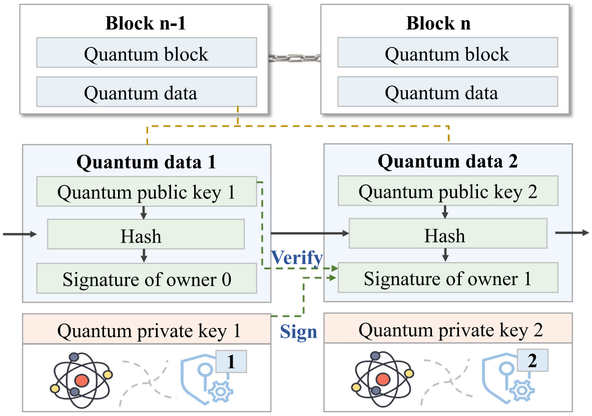

Fig. 5. ***Structure of quantum blockchain.***

on this novel method, more than one and a half nodes in the blockchain network sign the new block using the postquantum threshold signature. As a result, it has a higher level of security. Furthermore, much of the computational complexity of the consensus comes from the signature of the new block, which is much more efficient than the existing consensuses.

Compared to current threshold signature schemes (e.g., RSA-based and elliptic curve signature-based algorithms), postquantum threshold signatures are more promising and advantageous for future blockchain and other applications.

b) Vote-based consensus: Due to the node-independent computing power, the quantum capabilities of all parties are unbalanced, causing proof-based consensus to be inapplicable. Therefore, the vote-based consensus is more suitable for quantum blockchain in Web 3.0 [79].

The integration of a candidate block voting protocol, utilizing quantum threshold signatures, shows significant potential to enhance the robustness, security, and efficiency of blockchain systems. This approach offers superior fault tolerance compared to existing quantum consensus mechanisms [80]. To further promote fairness of vote-based consensus, a QDPoS based on quantum voting has been developed [81], allowing for fast decentralization even if the quantum computer emerges in the future. Similar to DPoS, QDPoS elects a certain number of representative nodes to generate new blocks by quantum voting. Quantum digital signature is introduced to ensure the quantum blockchain's efficiency and security. Based on QDPoS and quantum digital signature, a quantum blockchain scheme is designed, as illustrated in Fig. 5. Significantly, quantum blockchain uses quantum states to construct quantum blocks connecting by leveraging the entanglement property between quantum states.

As an extensively-applied consensus, PBFT is derived from the byzantine generals' problem, and its solution is the byzantine agreement protocol. To achieve byzantine agreement between multiple parties, a quantum byzantine agreement (QBA) without entanglement is proposed [82], including two stages. A group of semihonest distributors propagates a special list to the participants in the first stage via quantum secure communication (QSC). Afterward, the participants exchange information in the second stage to reach a consensus. The unconditional security of QKD contributes to the distribution of the relevant digital sequences, ensuring the security of the byzantine agreement protocol.

When designing a vote-based consensus, we must focus on its efficiency. To this end, a quantum honest-success byzantine agreement protocol introduces quantum protection and quantum certificate into the syntax of transactions [73]. It can break the bottleneck of certain quantum cryptographic protocols. Based on this consensus, the quantum-enhanced logic-based blockchain is constructed to improve its efficiency and power. Similar to the work in [82], there is no use of multiparticle entanglement in these quantum blockchains, making them easy to implement with existing technology.

B. Incentive Mechanism

While incentive mechanisms are well-studied in classical blockchain [83], the design of effective incentives for participants in quantum blockchain is still in its early stages within Web 3.0. To enhance network participation and reliability, incentives are used to motivate miner participation throughout the consensus process.

1) Quantum-Based Mining Strategies: In addition to classical PoW and PoS mechanisms in Web 3.0, miners in a quantum blockchain leverage quantum resources, e.g., QKD, to record new transactions and blocks, receiving tokens as rewards. For instance, Azhar et al. [84] apply quantum cryptography protocols, to ensure secure blockchain communication between participating nodes, protecting against third-party interference. Using the six-stage QKD protocol [85], the transparency and immunity of blockchain-based cryptocurrency systems can be ensured. Furthermore, by observing key rate generation on the numerical QKD simulation platform, the authors experimentally verify the feasibility of the protocol. This key generation protocol can be applied to any blockchain system as it is further developed.

Based on the need to set up additional infrastructures such as quantum servers, Bennet and Daryanoosh [86] propose the concept of PoE, which allows mining nodes of a quantum blockchain to reach consensus based on quantum resources. Specifically, an energy-efficient interactive protocol is proposed that allows quantum blockchain miners to execute formulas based only on their optically encoded quantum information without having to trust the network infrastructure. In the proposed quantum blockchain framework, a secure quantum authentication protocol is proposed to authenticate user clients, running devices, and communication links. Based on these protocols, sybil attacks and device tampering detection can be performed according to the rules of quantum mechanics without exposing user information. Finally, the authors discuss ways to explore existing frameworks, such as improving transaction throughput, more efficient consensus mechanisms, and reducing latency and memory consumption.

2) Secure Token Systems: Similar to the idea of reaching a consensus based on entanglement resources, Sun et al. [73] propose the incentive mechanism, namely, Qulogicoin, for the quantum-enhanced logic-based blockchain. This mechanism employs the byzantine agreement protocol to safeguard blockchain integrity against various malicious attacks, including double-spending, manipulation by adversaries, and interference from bribed distributors. Through implementation and testing with current technologies, the authors demonstrate that the proposed incentive mechanism is efficient and powerful.

In addition to the fungible tokens, NFT for quantum blockchain integrates key features of blockchain to create unique and authentic tokens with distinct properties. However, classical NFTs are often costly due to high energy consumption and security limitations. To address this, Pandey et al. [87] propose a new protocol for preparing quantum irreplaceable tokens, rather than making quantum states representing NFTs physically accessible to their owners. The protocol is ultimately more efficient than PoW and incorporates PoS. These quantum NFTs can be used to store patient monitoring and medical data cheaply and securely in the quantum blockchain. In this context, Kaushik and Kumar [88] propose an incentive mechanism for storing personal health data in the quantum blockchain that can ensure veracity, integrity, and availability.

C. Smart Contract

Existing smart contracts in Web 3.0 against quantum attacks strive to provide solutions for efficiency and scalability. On the one hand, quantum computing and quantum communication can provide demonstrable security in the development of smart contracts. On the other hand, smart contracts based on PQC are partially secure against quantum attacks. These two solutions can provide efficient and scalable solutions for signing, executing, and verifying smart contracts.

1) Quantum Cryptography-Based Smart Contracts: Cai et al. [89] propose a smart contract-based architecture that defends against quantum attacks using quantum blind signatures. Their analysis demonstrates that the quantum signature scheme, based on quantum entanglement, enhances blockchain smart contract security for both single and multiple signatures. Initially, they examine the message processing and transmission involved in this framework's quantum blind signatures. Then, they describe the lifecycle and rules of these signatures, explaining the design, operation, and security performance of the lightweight signature protocol. In addition, an advanced quantum blind signature algorithm for multiple signatures is introduced, along with a thorough analysis of its security. Furthermore, Cai et al. [90] propose a decentralized framework for multiparty transactions. This framework develops a quantum blind multisignature algorithm encompassing initialization, signing, verification, and implementation to offer computationally efficient and scalable solutions for quantum blockchain applications.

2) Postquantum Smart Contracts: PQC can enhance the resilience of smart contracts against quantum attacks in blockchain systems. For instance, Karbasi and Shahpasand [91] propose an end-to-end encryption scheme to protect against man-in-the-middle and eavesdropping attacks. In [92], a logic-based smart contract programming language, logicontract (LC), is developed for use in a quantum-secure blockchain framework. LC leverages modern declarative logic programming and incorporates postquantum signatures to address limitations in user protection on shared nodes. The next steps involve investigating the application of this logic language in secure multiparty computation and extending it to incorporate quantum logic, thus creating a programming language for smart contracts in quantum environments. As an important application in cloud computing, Li et al. [93] explore PQB for provable data ownership, aiming to mitigate the risks of single points of failure and partial trust associated with centralized auditors. Grid-based, privacy-preserving, verifiable data ownership is enabled based on smart contracts. The analysis verifies the correctness, soundness, and performance of the proposed system through a series of interactive games.

D. Postquantum Migration for Blockchchain

To effectively avoid the risk associated with prequantum blockchain, some postquantum migration methods have been proposed by many researchers in recent years. In this section, we explore the implementation of postquantum migration using the following mechanisms. We list details of the existing research works in Table 2.

1) Public Key Postquantum Cryptosystems: There exist four types of postquantum cryptosystems and a fifth type of cryptosystem with a mixture of prequantum and postquantum.

a) Code-based cryptosystem: It is based on the theory supporting error-correcting codes.

For example, McEliece's [111] cryptosystem can perform fast encryption and decryption in blockchain. Nonetheless, this cryptosystem requires storing and executing operations of large matrices, a restriction for resource-constrained devices. To address this problem, some code-based postquantum encryption cryptosystems investigate matrix compression techniques, as well as specific coding techniques. Notably, the National Institute of Standards and Technology (NIST) call for postquantum public-key cryptosystems is currently in its second round [94], [95], with the first standard draft expected to be released between 2022 and 2024. These cryptosystems adjust the

Table 2 Details of the Existing Postquantum Migration Methods for Blockchain in Web 3.0

Types	Ref.	Required Technology	Benefits	Contribution
Public key post quantum cryptosystems	[95]	Post-quantum cryptography	Security	Adjust the required security parameters of the cryptosystem
	[96], [97]	Square matrices with random quadratic polynomials	Higher decryption speed	Propose a new variant of the cryptosystem which is more suitable for Web 3.0 scenario
	[98]	Polynomial algebra	Security, efficiency	Propose a new public key cryptosystem, allowing for accelerated blockchain user transactions
	[99], [100]	Key-exchange protocol	Post-quantum security	Propose new parameters and a better suited error distribution to protect the exchanged data from quantum attacks
Post-Quantum Signature Algorithms	[101], [102]	Quantum digital signatures, quantum teleportation	Security, privacy	Propose secure mechanisms for performing transactions on the blockchain network
	[103]	Quantum-resisting signature	Low computational complexity, unforgeability	Present post-quantum blockchain and a secure cryptocurrency scheme to resist quantum computing attacks
	[104], [105]	Anti-quantum signature	Lightweight, security	Propose an anti-quantum transaction authentication scheme in the blockchain
	[106]	Post-Quantum Cryptography	Practicability	Develop a signature scheme for practicability and use in embedded systems
	[107]	Post-quantum cryptography	Internal technologies	Introduce a modified lattice-based signature scheme for decentralized PKI system
Quantum Networked Time Machine	[108]	Quantum entanglement	Scalability	Present a new approach generating quantum entanglement between many photons by using only a single source of entangled photon pairs to solve the scalability problem
	[11]	Quantum entanglement	Security	Overview of a conceptual design of a quantum blockchain using entanglement, which includes encoding the blockchain as a temporal Greenberg-Horn-Chalinger state of photons
Quantum Hashing	[109]	Quantum hash	Robustness	Present a novel multimedia identification system based on quantum hashing
	[110]	Fourier transform features, controlled randomization	Security, robustness	Develop a novel algorithm for generating an image hash

required security parameters while ensuring classical security between 128 and 256 bits. Compared with the current ECDSA [112] and RSA-based [113] encryption systems, these postquantum encryption cryptosystems offer the best tradeoff between security and key size.

b) Multivariate-based cryptosystem: It depends on the complexity of solving systems of multivariate equations, which turns out to be NP-hard [114].

Currently, some multivariate-based schemes are based on the use of square matrices with random quadratic polynomials, cryptographic systems derived from the Matsumoto–Imai algorithm [96], and schemes depending on hidden field equations [97]. These schemes are more suitable for Web 3.0 scenarios by reducing the key size and ciphertext overhead while increasing the decryption speed.

c) Lattice-based cryptosystem: It depends on the presumed hardness of lattice problems [115] that cannot currently be solved by quantum computers.

Lattice-based schemes allow for accelerated blockchain user transactions and can be executed efficiently as they are generally simple to calculate [116]. The implementations of these postquantum lattice-based schemes require the storage and utilization of large keys [98]. The general schemes guarantee classical security between 128 and 368 bits and quantum security between 84 and 300 bits. Based on the security level they provide, the complexity of these schemes varies significantly.

d) Hybrid cryptosystem: It merges prequantum and postquantum cryptosystems with the aim of protecting the exchanged data from quantum attacks and attacks against the postquantum scheme used. For instance, X25519 merging New Hope [99] with an ECC-based Diffie–Hellman key agreement scheme has completed the testing by Google [100].

Different from the existing public key infrastructure, the quantum state information cannot be copied. Thus, only those who know the exact quantum state can create and distribute public keys. Additionally, all the above cryptosystem implementations require significant computational resources and more energy consumption. Therefore, the postquantum cryptosystems for blockchain should aim to find a balance between security, computational complexity, and resource consumption.

2) Postquantum Signature Algorithms: The rapid development of quantum computers poses significant security risks to the cryptographic algorithms underlying blockchain networks. To overcome the vulnerabilities of the blockchain networks to quantum adversaries, some potential antiquantum signature algorithms have been proposed.

Using quantum digital signatures and quantum teleportation phenomenon, a quantum secure theme is presented to implement the blockchain [101]. In particular, messages are signed by quantum digital signatures [102] and then they are distributed across the blockchain network by the teleportation phenomenon, creating the opportunity to build a more secure and faster blockchain for Web 3.0.

Based on lattice cryptography, novel quantum-resisting signature schemes are used for transaction authentication in blockchain-enabled systems. In particular, the lattice basis delegation algorithm provides a way to generate secret keys [103]. Then, the double signature, defined as the first signature and last signature, is developed to reduce the correlation between the message and the signature. The secure cryptocurrency scheme combines the proposed signature scheme with the blockchain, satisfying correctness and having the advantage of resisting quantum computing attacks.

Different from traditional elliptic curve signatures, a new signature authentication scheme leverages Bonsai Trees technology to generate the extended lattice accompanied by the corresponding key and then generate the signature [104]. This signature enables a robust, unforgeable level of security for antiquantum transaction authentication, even under selected message attacks. By combining Bonsai Trees technology and the RandBasis algorithm, public and private keys can be generated for transaction verification [105]. Notably, this scheme produces smaller public and private keys and signatures, reducing computational complexity and enhancing implementation efficiency. Güneysu et al. [106] present a lattice-based signature scheme optimized for embedded systems, requiring a 12 000-bit public key and a 2000-bit private key to generate 9000-bit signatures for a 100-bit security level. Due to its simplicity and efficiency, this scheme is considered a promising signature algorithm for managing public key encryption for a postquantum decentralized system, such as QChain [107].

3) Quantum Networked Time Machine: It is a conceptual design for quantum blockchain. This means if a quantum blockchain is to be constructed, it could be regarded as a quantum networked time machine.

Entanglement in time, as opposed to entanglement in space, plays a key role in the quantum benefits of classical blockchains. Megidish et al. [108] use the temporal greenberger-horne-zeilinger (GHZ) state of the photon as a blockchain, providing a key quantum advantage over spatial entanglement. In this conceptual system, a temporal Bell state as a block contains two classical records, and a growing temporal GHZ state has been taken as a chain.

In this way, the functionality of timestamped blocks and hash functions are replaced with temporal entanglement [11]. The quantum advantage is a significant amplification of the susceptibility to tampering, meaning that if one user tampers with a single block, the complete local copy of the blockchain might be destroyed. While for a classical blockchain, only the block after the tampered block is destroyed, making it vulnerable to vulnerabilities. This work offers the realistic possibility of deploying pure quantum blockchains.

4) Quantum Hashing: It aims to improve the robustness of the binary hashing systems using the same intermediate hash values against various distortions.

Based on the severity of the distortion, the Hamming weight of the D-bit binary hash difference between a query and its true-underlying content can vary from 0 to D. The probability that each bit of the D-bit binary hash difference evaluated by the quantum hashing system is 1. Instead of using a definite hash value in a binary hash, quantum hashing incorporates uncertainty in the hash value [109]. More importantly, the main difference between a quantum hashing system and a binary hashing system is the way the dissimilarity between the query and the candidate data in the binary hash database is calculated.

In the binary hashing system, the system extracts the intermediate hash from the query, encoding it as a binary hash to locate a match within the binary hash database [110]. Then, the dissimilarity between the query and the candidate data is obtained. Unlike binary hashing, the quantum hashing system does not require additional storage as the quantum hashing algorithm uses the same binary hash database as the binary hashing algorithm. Additionally, acceptance and rejection decisions can be made with a slight increase in computational cost. It provides a novel postquantum migration algorithm, enhancing blockchain resilience against quantum attacks.

a) Lessons learned: Optimizing internal technologies in quantum blockchain has shown that quantum-based consensus protocols, such as entanglement, can enhance security against quantum threats. Quantum-enabled incentives, such as proof-of-entanglement, improve transaction security and participation while integrating quantum cryptography into smart contracts to strengthen defenses despite scalability challenges. Seamless migration to quantum-safe algorithms is vital for blockchain integrity, requiring secure and compatible systems. Balancing security with resource efficiency is also key for sustainable, scalable quantum blockchain systems.

VI. EXTERNAL PERFORMANCE IMPROVEMENT OF QUANTUM-DRIVEN WEB 3.0

A. Quantum Blockchain Economy

The quantum blockchain economy refers to establishing a secure and decentralized Web 3.0 economic model with the support of quantum and blockchain technology. In this section, we mainly explore the quantum blockchain economy in the following two aspects: the payment protocol and the market mechanism. We list details of the existing research works in Table 3.

1) Payment Protocol: E-commerce has become a major force in the global economy because of the convenience and accessibility it provides. The growth of E-commerce

Table 3 Details of the Existing Works in External Performance Improvement

Type	Ref.	Quantum Technology	Benefits from quantum	Contribution	Security Analysis	Storage Analysis
Quantum Blockchain Economy	[118]	quantum key distribution	Unconditional security	Propose a new E-payment system based on quantum blind and group signature	✓	×
	[119]–[121]	Quantum proxy blind signature	Security, anonymity	Develop e-payment systems to support inter-bank transactions	✓	×
	[122], [123]	Quantum proxy blind signature, quantum key distribution	Security, anonymity	Describe in detail the implementation process of the quantum e-payment system	✓	×
	[124]	Quantum computation	Anonymity,verifiability, eligibility	Develop a simple voting protocol based on quantum blockchain	✓	×
	[125]–[127]	Quantum communication	Security, fairness	Present the economic and feasible quantum auction protocols	✓	×
	[128], [129]	Quantum bit commitment	Unpredictability, unforgeability, verifiability, decentralization	Present a protocol for auction on quantum blockchain	✓	×
Decentralization	[130]	Quantum computing, quantum networks	Decentralization, security	Provide a theoretical analysis of the quantum blockchain for decentralized identity authentication	✓	×
	[131]	Quantum identity authentication, zero-knowledge proof	Unconditional security	Propose a theoretical scheme for zero-knowledge proof quantum identity authentication	✓	×
Scalability	[132]	Quantum networks	Decentralization, scalability	Integrate smart contracts and quantum lightning to improve the speed of payment transactions	✓	×
	[133]	Quantum teleportation	Low throughput and latency	Propose a consensus mechanism to reduce resource consumption	×	✓
Privacy of storage	[134]	Post-quantum encryption	Low storage cost, robustness	Propose a secure threshold verifiable multi-secret sharing scheme for transaction verification and private communication	✓	✓
	[135], [136]	Quantum teleportation	Low throughput and latency	Present a secure threshold-based verifiable multi-secret while improving the recovery communication cost and robustness	×	✓
	[137]	Quantum signature	Anonymity, unforgeability, shareability	Develop an anti-quantum signature for secure data sharing with blockchain	✓	×

has also led to the development of new payment models, such as electronic cash (E-cash) payment protocols. Compared with other protocols, E-cash is becoming a more desirable payment protocol due to its anonymity and offline transferability. Currently, the security and cryptography of electronic payment have been the focus of research. Particularly, classical cryptography in electronic payment might be threatened by quantum computers with the development of quantum computing [117].

To address the above challenges, quantum cryptography, combining classical cryptography and quantum theory, appears to ensure the unconditional security of payment protocol. Based on the quantum blind group signature with two trusted third parties [118], an e-payment system is established, and then, a cross-bank electronic payment protocol based on the quantum proxy blind signature emerges [119]. Nevertheless, the presence of a dishonest customer in a payment protocol leads to a risk that some purchase information may be leaked. Different from the proxy blind signature, quantum multiproxy blind signature and group blind signature can be used in the e-payment protocol with a third-party platform to further guarantee the customer's anonymity [120], [121].

Blockchain acts as a trusted third party, integrated with quantum technology, to achieve reliable trust between both parties of the transaction [122], [123]. These studies propose an e-payment protocol implemented by blockchain and quantum signature. Specifically, these works describe in detail the main characters and implementation process of the quantum e-payment system as follows.

a) Main characters: Five characters exist in the quantum e-payment system: 1) customer; 2) two customer agent banks, denoted as bank A and bank B, respectively; 3) representative of the bank, denoted as R; 4) merchant; and 5) blockchain.

b) Implementation process: When a customer pays for online purchases, he finds that bank A has a low balance. In this case, he runs out of bank A's money, and then, he pays bank B the rest of the price. In this process of transaction, R receives the purchase information of the customer by QKD protocols. Meanwhile, R plays as the proxy signer to sign the payment slip when the settlement is completed. After, the customer sends a signed blind copy to the merchant by using the BSM performed by agent banks A and B.

By leveraging the property of quantum proxy blind signature, the anonymity of e-payment is guaranteed. Furthermore, blockchain technology stores all the information of customers and helps the merchant verify the legitimacy of the signature. This scheme has good application prospects and wide application value.

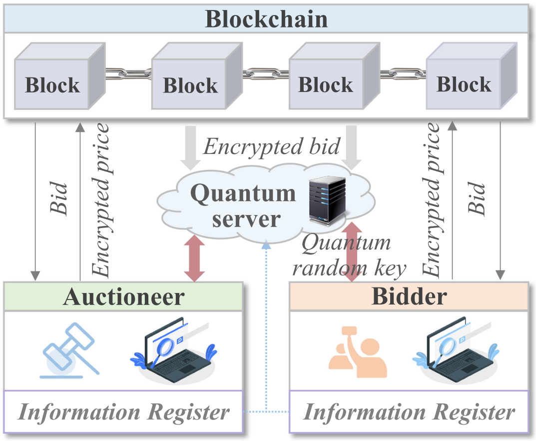

Fig. 6. *Auction on the quantum blockchain.*

2) Market Mechanism: Currently, quantum-enabled blockchain focuses on specific market mechanisms, such as voting, lottery, or auction protocols.

A number of voting protocols based on quantum blockchain have been developed to address the risks posed by the upcoming quantum computers, providing anonymous, verifiable, fair, and self-tallying voting while simplifying the task of electronic voting (E-voting). Similar to the voting protocol on the Bitcoin blockchain, the voting protocol supported by the quantum blockchain consists of two phases: a ballot commitment phase and a ballot tallying phase [124]. In this voting protocol, QSC is leveraged to distribute the matrix by voters, and then, voters give the masked ballots to miners. Afterward, QBA is used by miners to achieve consensus about voters' masked ballot.

Moreover, auctions are one of the more important market mechanisms in the quantum blockchain economy. It allocates resources according to a set of market rules and prices determined by buyers' bids. Blockchain-based auctions [125] satisfy decentralization in auctions, and quantum auctions [126], [127] satisfy unconditional security, but existing auction protocols cannot guarantee both properties. In [128], an auction on the quantum blockchain is proposed to solve this problem. Fig. 6 shows a simple flow of the auction process. There are three types of participants in this auction protocol, sellers, buyers, and miners. This auction process is divided into five stages.

1) *The bidding phase:* Each buyer announces his bid to the sellers and all miners by qubit commitment.
2) *The opening phase:* Each buyer opens his bid to the sellers.
3) *Decision phase:* The winning bid and the winning buyer are determined.
4) *Verification phase:* This stage convinces miners that the buyer has picked an effective winner.
5) *Publication phase:* Achieving consensus and adding it to the blockchain.

In general, the transactions of the auction are stored in the blockchain and supported by quantum computation and communication to enhance security and protect privacy. As one of the major auction protocols, the sealed-bid auction protocol using a quantum-based blockchain is explored [129]. This protocol uses quantum computing and blockchain technology to guarantee essential features and requirements.

1) *Registration:* Bidders interested in participating in the auction register their personal information on the quantum server.
2) *Generating quantum keys:* Quantum server shares random QKD-based keys with the auctioneer and bidder to encrypt data and authenticate their identities.
3) *Data encryption and submission:* Each bidder and the auctioneer encrypt the private bid and the lowest acceptable price using the encryption keys and submit the encrypted data to the blockchain, respectively.
4) *Data evaluation and announcing the winning bid:* The quantum server downloads all submitted bids and the auctioneer's encrypted data to calculate the winning bids and submit them to the blockchain.
5) *Verification:* Each bidder verifies the winning bid by comparing the submitted bid with the winning bid.

The quantum blockchain-based lottery and auction protocols satisfy the significant features of distributed lotteries and auctions and can be implemented by existing technologies. It is demonstrated that quantum blockchains can provide new insights into its market economic mechanisms.

B. Decentralization

Decentralization and transparency can enable blockchain technology to gain widespread attention across a wide range of applications. Accordingly, the decentralization of quantum blockchain is also being explored.

In the quantum blockchain, classical data chains are redesigned in a quantum way by associating quantum states with entanglement. In this case, quantum decentralization is claimed to be "unconditionally secure." Moreover, quantum blockchain infrastructure, such as access control and user authentication, must be established prior to any quantum decentralization. In this article, we focus on decentralized identity management in the quantum blockchain.

The current Internet relies on centralized storage for users' digital identities, which compromises user privacy and faces vulnerabilities such as single points of failure or data tampering. To address these issues, decentralized identity management based on quantum blockchain has been proposed. The quantum blockchain identity framework (QBIF) is a popular method for creating secure pseudonyms [130]. Within QBIF, users retain control over their identities and can interact without disclosing unnecessary information. This framework ensures identity security and authenticity in a decentralized quantum environment, offering enhanced privacy protection and preventing forgery [138], [139]. Next, we mainly introduce various aspects of the QBIF architecture, covering roles, quantum identity attestations (QIAs), and quantum blockchain.

1) Roles in QBIF: Within this framework, QBIF has three roles, including identity owners, identity issuers, and identity verifiers. Identity issuers represent trusted parties that issue identities. In addition, they verify the validity of identity attributes and issue attestations, stored on blockchains. Users, i.e., owners, determine their own identity and use on-chain authentication to prove to verifiers their identity. The verifiers provide services based on the reputation of the issuer who offers the attested identity.

2) Quantum Identity Attestations: QIAs are stored as quantum states and chained together by entanglement [140]. The QIAs are generated by issuers, and then verifiers perform QIA authentication according to user requirements. After, the users who want authorized actions from verifiers need to perform the authentication and proof process.

Attestations are evidence of statements about the user's identity, and they can be hash values, encrypted messages or bit strings, etc. QIAs in QBIF can be created by converting them into qubits [131]. A user can have multiple QIAs for different identity properties. Before using the QIA, it should be double-signed by the owner and the issuer, which are stored in the chain of quantum states. The signature is done between them using the private key signature of both parties. The use of the issuer's private key signature illustrates that the issuer agrees with the user's attested statement, while the use of the user's private key signature indicates ownership of the attestation. No trusted centralized storage is required.

3) Identity Management on Quantum Blockchain: QBIF provides distributed identity management for coordinating quantum network nodes (including users, issuers, and verifiers). These nodes are interconnected through a quantum communication channel. Each type of quantum network node has a copy of a quantum state chain that holds signed QIAs in chronological order. Once a QIA is kept in the blockchain, the user sends the location of the signed QIA to initiate the verifier's operation. Then, the verifier validates the signed QIA using the public keys of the user and the issuer to determine the authenticity of the QIA. It is worth noting that the validity level of the validity can be defined by the issuer using the password, online, etc. In addition, the validity level can be integrated into the QIA by adding additional qubits.

This quantum blockchain identity management deals with the primitive quantum blockchain structure, where some data transfer and payment transactions can take place off-chain. With the development of quantum technology, a hybrid network of classical and quantum communications might be the best way to transition from classical to quantum Internet. Meanwhile, the proposed framework faces some limitations and challenges, but it shows an obvious path to realize the decentralization of quantum blockchain.

C. Scalability

Scalability is a central issue of quantum blockchain that has attracted researchers' attention. It aims to increase transaction throughput (i.e., transactions per second) while keeping the resources required for a party to participate in the consensus mechanism roughly constant and maintaining security against adversaries.

Traditional blockchains, such as Bitcoin and Ethereum, are currently constrained to processing around ten transactions per second, highlighting significant scalability limitations. Quantum information technology offers a promising solution to this challenge. By integrating quantum cryptography, classical blockchains can potentially solve scalability limitations. Furthermore, smart contracts can be enhanced with quantum tools, specifically quantum lightning, to establish a decentralized payment system that addresses the scalability of transactions. Quantum lightning, introduced by Zhandry [141] and inspired by collision-resistant quantum money [142], features a payment system that involves generating and verifying quantum banknotes. This method assigns unique serial numbers, ensuring security by making it statistically improbable for any process to produce two identical banknotes. This property necessitates a protocol to control the issuance of legitimate quantum banknotes, thereby ensuring transaction integrity.

Coladangelo and Sattath [132] utilize smart contracts to manage serial numbers in quantum transactions, allowing any party to deposit coins into a contract, thus assigning value to a quantum banknote. This integration between smart contracts and quantum lightning improves the speed of payment transactions, further tackling the scalability issues. Additionally, a consensus mechanism utilizing quantum cryptography, teleportation, and measurement has been developed to reduce throughput and latency issues [133]. This approach assesses miners' success in mining without requiring extensive computational resources or complex calculations, thereby minimizing resource consumption. Furthermore, quantum teleportation mainly consists of classical information transmission and quantum gates performing. It takes a very short time, while the transmission of quantum states is completed instantaneously, contributing to high scalability and low theoretical delay in this scheme.

D. Privacy of Storage

Traditional blockchain systems store transaction data in the form of a distributed ledger, with each node storing copies of all data, which raises the storage problem. With the development of quantum technology, numerous researchers are looking for suitable quantum/postquantum encryption and verification algorithms to improve standard distributed storage blockchain systems [143], [144], [145].

1) Postquantum-Based Distributed Storage: Existing research work proposes some secret sharing schemes to overcome the problem of distributed storage. Secret sharing, as one of the most important encryption protocols, is used to ensure the privacy of data [146], [147].

A secure threshold-based verifiable multisecret sharing scheme is proposed that does not require a private channel and shares two secrets simultaneously in a single sharing process among blockchain nodes based on postquantum encryption algorithms [134]. The scheme is then applied to a distributed storage blockchain system for the distribution and private storage of data blocks. Before distributing the data among the nodes of each block, we encrypt the data blocks with the AES-256 encryption algorithm. Meanwhile, this scheme shares secret keys and hashes of the blocks among the blockchain network nodes simultaneously. After, the encrypted data blocks are encoded by Reed–Solomon codes and shared among these blockchain nodes.

In addition, several secret sharing schemes based on lattices have been studied in [148] and [149]. Based on these postquantum encryption schemes, the distributed storage blockchains also greatly improve the storage costs of traditional blockchains [135], [136]. The scheme also brings desirable features to distributed storage blockchain systems, such as (quantum-secure) authentication algorithms and secret communication without private channels. Notably, the flexible threshold parameters of the proposed scheme eliminate the shortcomings of the previously distributed storage blockchain in terms of recovery communication cost and robustness.

2) Implementation Tutorial for Storage: Taking EMRs as an example, this section introduces how to realize distributed data storage with blockchain-based quantum technology.

Along with the unceasing progress of medicine, it has become a popular trend to implement EMRs to improve the efficiency and reliability of medical services [150], [151]. Nonetheless, EMRs are stored independently in hospitals and healthcare facilities, which leads to sharing problems. In addition, highly sensitive EMRs are vulnerable to being tampered with, posing a threat to privacy and security. An antiquantum attribute-based signature (AQ-ABS) is developed to address the above challenges [137]. AQ-ABS combines IPFS with the consortium blockchain to encrypt and sign EMRs, and then store them in a secure and distributed file storage system, i.e., IPFS. After, the index hashes generated by IPFS and keywords are resigned and stored in the blockchain. As shown in Fig. 7, the system implementation process for EMR shared plans is divided into four phases.

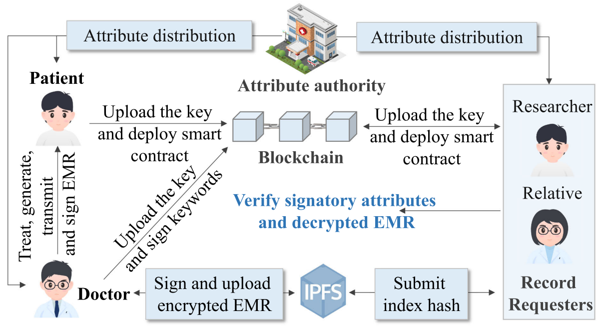

Fig. 7. *System architecture of AQ-ABS.*

1) *System initialization:* This step includes registration and security parameter settings. These attributes and secret keys are uploaded to the blockchain.
2) *Electronic medical record (EMR) generation:* When a user is treated for a disease at one hospital, it can consult the hospital and generate an EMR, as in Fig. 1. If the user needs to be transferred to another hospital for treatment, the physicians at that hospital must be familiar with the patient's previous medical experience. That is, each physician is responsible for uploading the EMR for data sharing. In this case, the EMR record is generated during the doctor's treatment of the patient.
3) *Signature generation and uploading:* The physician responsible for the consultation sends an EMR to the patient to confirm the accuracy of the EMR. After receiving the EMR with the user's signature, the physician encrypts the EMR by Brost et al. [152] and signs it to ensure unforgeability based on AQ-ABS. In addition, a combination of IPFS and consortium blockchain is used to achieve secure EMR sharing. A physician uploads an encrypted EHR with a double signature to IPFS to obtain an index hash. The physician, then, signs the index and keywords of the EHR and uploads them to the blockchain for distributed storage, while the patient deploys a smart contract to the blockchain for access control of the EHR.
4) *Query and signature verification:* This step allows the user to download the EMR and verify the signature. As a patient's relevant researchers wish to access the EMR, they first provide their attributes to the blockchain, which is verified by the smart contract, and search the index. If it is accessible, the blockchain returns the index value with the signature; otherwise, it rejects it. After receiving the signed index, the record requester verifies the signature, and if it passes, the verification submits the index to IPFS for EMR.

a) Lessons learned: Enhancing external performance in quantum-driven Web 3.0 reveals promising advancements. Quantum-based payment protocols and market mechanisms enable secure, decentralized, and scalable economic models. Quantum technologies improve

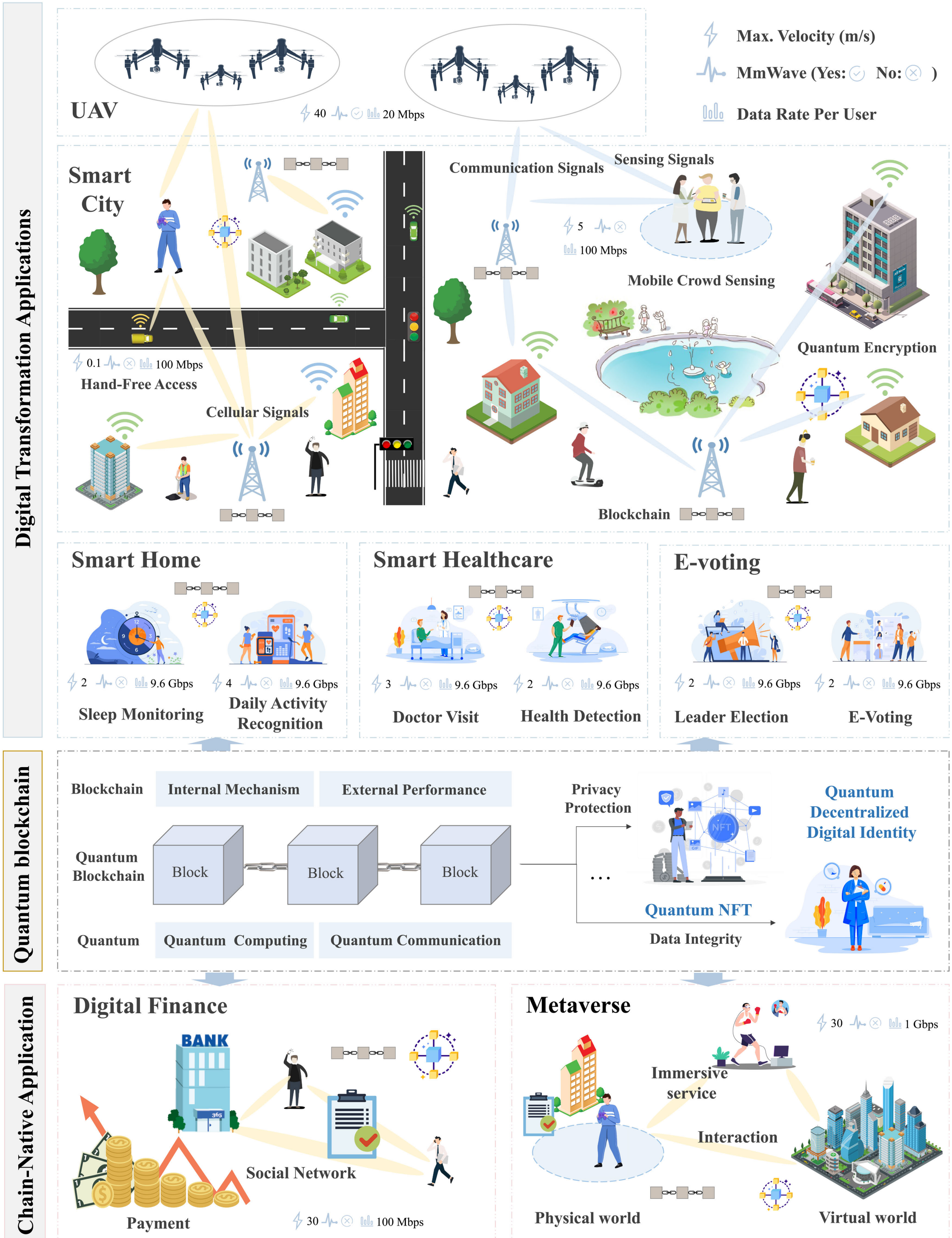

Fig. 8. ***Applications of quantum blockchain in Web 3.0.***

decentralization by reducing vulnerabilities and strengthening user privacy and data security. Additionally, quantum and postquantum encryptions enhance data privacy and storage scalability, supporting secure data management in Web 3.0. These advancements highlight the quantum blockchain's role in building a scalable, secure, and decentralized digital ecosystem.

VII. APPLICATIONS AND IMPLEMENTATION TUTORIALS OF QUANTUM BLOCKCHAIN IN WEB 3.0

This section briefly introduces some applications and implementation tutorials of quantum blockchain in Web 3.0. These applications include chain-native services and digital transformation applications, as shown in Fig. 8.

A. Chain-Native Services

1) Digital Finance: Modern digital finance prioritizes high reliability and security. Leveraging digital ledger technology, quantum blockchain can store vast amounts of financial data on a decentralized network. As a significant blockchain-based innovation, NFT is a nonfungible digital asset that cannot be replaced by its equivalent one [153]. Further, it bridges the physical and digital worlds through a range of applications, including identity verification, private equity transactions, intellectual property management, and digital collections. To ensure the security of quantum NFT, the protocol proposed in [87] introduces entanglement to replace traditional ledger and hash functions, enhancing blockchain security in a quantum computing context.

This quantum NFT protocol generally has three steps.

1) *Creation of block of NFT:* The quantum NFT protocol takes a Bell state, where the first qubit of the Bell state contains information about the owner, while the second qubit stores information about the token.
2) *Creation of token:* A Hadamard gate is leveraged to place a qubit in a superposition of zero and one, further creating a token that is a random and unique phase.
3) *Verification of the blocks:* This step utilizes the entanglement of double-weighted hypergraph state to ensure that blocks are added by consensus, and to perform verification.

2) Metaverse: Based on DeFi, metaverse operates with its own economy, drawing escalating attention to the next-generation Internet. Generally, metaverse is defined as a shared virtual space formulated by sensing, communication, and computing technologies. Meanwhile, other technologies, such as blockchain and quantum computing, are proving popular for a number of key metaverse applications [154], as shown in the following.

1) *Security:* As more interactions between the physical and the virtual world in the metaverse are captured, quantum-resistant blockchain technology is designed for all transactions and commerce. This also ensures that blockchain transactions remain safe against algorithms such as Shor's algorithm [155].
2) *Computation:* One of the significant advantages of quantum blockchains is that they are computationally efficient, ensuring the increased effectiveness of metaverse applications. Concretely, quantum blockchain enhances the computation and overall experience, which makes the metaverse user-friendly.
3) *Randomness:* To ensure that the blockchain-based metaverse system is not manipulated by occupants and algorithms, a certain amount of quantum randomness is needed. Instead of pseudorandom numbers, a qubit series is leveraged to generate random bits, further preventing the metaverse from being utilized in unscrupulous ways. Currently, some companies, such as quantum dice [156], are working in the field of quantum random number generation.

B. Digital Transformation Applications

1) Smart Cities: Smart cities aim to enhance daily life, but IoT devices and centralized data systems are vulnerable to cyberattacks and privacy issues. While blockchain provides secure, decentralized data management, quantum computing could undermine the encryption that blockchains depend on.

In this regard, quantum cryptography helps to eliminate the risk of data storage and transmission associated with blockchain and IoT [157]. Abd El-Latif et al. [158] design a novel authentication and encryption protocol using quantum-inspired quantum walks (QIQW). The QIQW protocol enables IoT nodes in smart cities to securely share data and have full control of their records. Meanwhile, the novel blockchain framework based on QIQW can resist probable quantum attacks.

In addition, PQB is another solution to improve the security of transaction processing in the blockchain. To construct a lightweight PQB transaction in a smart city, Chen et al. [159] embed a postquantum identity-based signature scheme and the IPFS. Some postquantum solutions [160], such as lattice-based cryptography, QKD, and entanglement, are experimentally easy to counter quantum attacks, further creating a PQB-based smart city. This PQB architecture makes smart cities safer and provides a better place to live.

2) Smart Healthcare: Smart healthcare aims to deliver patient-focused services through secure data collection, efficient processing, and knowledge extraction [161]. Traditional systems struggle with privacy and security, but blockchain can improve both the efficiency and security of healthcare [133].

The advent of quantum computing introduces the risk that certain malicious medical entities could monopolize block mining in blockchain systems [162]. Naturally, developing a quantum blockchain and quantum EMRs (QEMRs) that leverage the inherent security of quantum cryptography would be highly beneficial [137]. To protect traditional healthcare systems from quantum attacks, the quantum blind signature is leveraged for key distribution during the block creation [133]. Smart contracts enable users to access blockchain-based medical data according to their designated roles. By replacing traditional encryption algorithms with quantum authentication, a QEMR protocol has been designed [163]. This protocol securely tracks every medical record, ensuring the privacy and integrity of EMRs in healthcare systems.

Additionally, quantum computing can support large operations and complex computation on uninstructed data, further improving the quality of service provided by intelligent healthcare systems [164], [165]. The work in [166] presents a quantum cloud-as-a-service for secure medical computations. Specifically, blockchain enables secure communication between various medical members. Delegated quantum cloud is utilized to securely compute medical data while protecting information privacy. To further perform accurate and efficient diagnoses, a more lightweight model integrating convolutional neural networks and quantum kernels is proposed [167]. Meanwhile, SHA-256 algorithms are utilized to encrypt and decrypt MRI images to ensure the integrity of medical information.

3) E-Voting: E-voting is a voting process in which votes are cast and counted with computer assistance. Nevertheless, any flaw in these E-voting protocols would allow malicious results to be manipulated. And with the development of quantum computers, these protocols might become even less secure.

Recently, the possible integration of blockchain and quantum technology could facilitate the implementation of E-voting with much greater transparency and security [168]. Sun et al. [124] propose a voting protocol based on the quantum blockchain. This protocol not only simplifies the task of E-voting but also satisfies many significant properties, such as anonymity, verifiability, eligibility, and fairness. However, this protocol is limited in that it cannot be applied beyond a certain number of users. In addition, denial-of-service attacks can easily be used by one voter against another. To mitigate potential attacks from emerging noisy intermediate scale quantum (NISQ) computers, an anonymous voting scheme has been developed by leveraging the advantages of blockchain technology enhanced with quantum protocols [169], such as quantum random number generators and quantum key distribution (QKD). This approach strengthens security and ensures resilience against the potential threats posed by NISQ computers. The quantum protocols enable the E-voting systems to secure against any adversary, whereas there exists the central authority in the elections, which may adversely affect the voting outcome.

Hence, it is necessary to consider the audit function in the E-voting protocol in order to ensure the efficiency and fairness of the voting process. Gao et al. [170] design a novel E-voting protocol, providing transparency in the voting environment. Different from the cryptosystem based on the difficulty of number theory, this protocol combines the certificate traceable ring signature in [171] with the code-based algorithm. It realizes the anonymity of voters and has the ability to audit voters operating incorrectly, further resisting quantum attacks.

4) Smart Home and UAV: As prominent representatives in the IoT, smart homes and UAVs have surged forward in the past decade, improving our daily lives.

On the one hand, smart homes aim to offer intelligent sensing and collaborative control in the home. It is noted that security is an important problem before it can be widely adopted. One plausible solution to security breaches is to introduce blockchain and quantum technology to increase safe communication among smart devices [172]. To this end, a consortium blockchain can be formed to provide smart home services through smart contracts. Meanwhile, the postquantum signature schemes, such as pqNTRUsign, are designed to replace existing ECDSA signature schemes, helping to protect the blockchain-based smart home services from a potential quantum computer attack.

UAVs, on the other hand, are seen as a potential asset due to their broader application in scope [173]. A number of communication protocols are needed to support the successful flight of UAVs. In UAV communications, blockchain-based quantum systems could be developed to solve security issues in the distribution of critical information. Thereinto, a quantum cryptography-based architecture leverages the properties of quantum cryptography and beyond 5G networks to enable drone communication more shielded from data transmission and to achieve high security and efficient data transmission.

C. Existing Quantum Blockchain Platforms for Web 3.0

Blockchain is an ideal platform for implementing advanced quantum-resistant cryptographic methods. Many platforms are leading efforts in quantum computing within the blockchain sector, exploring its potential impact on the technology.

International business machines corporation (IBM) has been at the forefront of quantum computing research and development, offering the IBM quantum platform [174], a cloud-based service that provides both public and premium access to quantum computing resources. Through this platform, IBM drives important advancements in quantum-resistant cryptography, specifically aimed at securing blockchain systems against future quantum threats. Similarly, Rigetti computing [175], known for its quantum cloud services and hardware development, provides quantum computer access through its cloud platform and is actively researching quantum cryptography to enhance blockchain security. Meanwhile, Alibaba has entered the quantum computing arena with Alibaba cloud quantum [176], providing cloud-based access to quantum resources and exploring quantum-resistant cryptography for blockchain applications.

QANplatform [177], the world's first quantum-resistant private blockchain compatible with Ethereum's EVM [178], is revolutionizing Web 3.0 operating systems and accelerating blockchain adoption. It achieves this by enabling a lattice-based postquantum cryptographic algorithm, supporting multilanguage smart contracts, and offering rapid cloud platform deployment. By integrating the postquantum algorithm recommended by the NIST into its quantum-resistant private blockchain, QANplatform has significantly enhanced its defense against quantum computing threats. Additionally, SQE [179], a quantum-secure decentralized blockchain platform powered by simulated quantum entanglement technology, is designed for industrial, commercial, and consumer applications. Its custom entangled hardware generates unique, nontransmitted encryption keys for every transaction, providing top-tier security against quantum computing threats.

D. Implementation Tutorials of Quantum Blockchain in Web 3.0

In this section, we provide a tutorial on the implementation of quantum blockchain in Web 3.0, which consists of the following four phases.

1) Generation of Quantum Entropy and Postquantum Certificates: In cryptographic processes in quantum blockchain, sufficient randomness prevents a large number of malicious attacks. The generation of quantum randomness takes advantage of the nondeterministic property of quantum mechanics [180], [181]. Considering the distributed nature of the blockchain, ideally, each blockchain node should have its own local source of quantum entropy. Then, quantum communication protocols need to be designed so that blockchain nodes create a quantum secure tunnel between themselves and the entropy distribution point to ensure secure communication. In this respect, the entropy source creates the first key, splits it into parts, and delivers it to the blockchain node through various transport layer security (TLS) channels.

Once a blockchain node accesses quantum entropy on demand, the entropy is consumed by OpenSSL, the cryptographic framework for applications that use TLS/secure sockets layer (SSL). After that, the blockchain nodes encapsulate the communication with other nodes using the postquantum key generated by the signature algorithm and sign the transactions broadcasted to the blockchain [95].

2) Nodes Communication Based on Quantum Cryptography: Communication between nodes is established through quantum protocols in blockchain, making nodes' communication quantum-resistant. It can be achieved by adding peer-to-peer TLS tunnels that have been modified to support postquantum keys. Since this is built on a TLS connection that is insensitive to key length, other postquantum algorithms can be used, unlike blockchain transactions [182].

3) Signature of Transactions: The implementation of this phase requires us to pay attention to each specific blockchain network. Some blockchain protocols recognize different encryption algorithms or have the flexibility to incorporate new ones. In this case, we need to add quantum signatures to transactions broadcast to the network without modifying the blockchain protocol by developing relay signers and meta-transaction signature models [183].

4) On-Chain Verification: When a blockchain writer node adds a postquantum signature to a meta-transaction and broadcasts it to the network, it needs to be verified on-chain. Once a relay hub smart contract is verified as a transaction target, each node extracts the original transaction information to verify the signature. The process involves three steps.

1) The blockchain node receiving the meta-transaction checks the sender. Next, they verify that the public key derived from the meta-transaction signature controls the DID of the node that generated the transactions [184].
2) If the previous step completes successfully, the node invokes the DID registry again, after which it resolves the postquantum public key associated with the DID and the blockchain public key verified in the previous step.
3) After, the postquantum public key, postquantum signature, and original transaction are obtained from the DID registry. Further, each node verifies the postquantum algorithm on-chain.

With the above steps, blockchain nodes add meta-transactions to the transaction pool and copy them to other nodes so that the validator receives them and adds them to the next block.

a) Lessons learned: Practical applications and implementation methods play a vital role in advancing quantum blockchain within Web 3.0. Areas such as digital finance, the metaverse, smart cities, and secure e-voting exemplify how quantum blockchain can enhance security and reliability in service delivery. Key frameworks highlighted in the implementation tutorials, including quantum entropy generation and QSC, underscore the need for adaptable, layered approaches for successful deployment. These insights offer a clear pathway for future advancements and adoption in Web 3.0.

VIII. FUTURE RESEARCH DIRECTIONS AND OPEN ISSUES

Although blockchain holds great promise in Web 3.0 with the help of quantum information technology, there are ongoing challenges in future research. In this section, we discuss several open issues and explore future directions, including quantum crypto-economics, quantum intelligence, and transaction intelligence.

A. Migration to Quantum Blockchain and PQB

Considering that classical cryptography is expected to become vulnerable, it is essential to ensure a smooth and efficient migration to quantum and PQC-based systems. For example, future advancements in quantum computers could threaten wallet security, especially for dormant accounts, underscoring the need for quantum-resistant cryptographic solutions. In this process, the legacy blockchains of Web 3.0 also need to be migrated to quantum blockchain and PQB. However, this migration is not just about adding QKD and postquantum cryptographic algorithms to existing blockchains.

To enhance quantum resistance and wallet security, Ethereum co-founder Vitalik Buterin has proposed solutions such as account abstraction and new transaction types [185]. Account abstraction would protect private keys in smart contract wallets using technologies such as STARKs and Winternitz signatures, while the new transaction type enables the use of quantum-resistant signatures. Moreover, the Ethereum community is exploring quantum-safe algorithms to activate a fault safety fork, aiming to safeguard blockchain security against future quantum computing risks.

B. Standardization for Quantum Blockchain and PQB

Standardizing quantum blockchain and PQBs depends on establishing unified standards for quantum networks and PQC. Once standardized, Web 3.0 can be scaled up for large-scale industrial deployment, significantly enhancing the scalability and service diversity of digital ecosystems. Standardizing quantum networks can pave the way for efficient and highly secure cryptographic schemes tailored for quantum blockchains.

Additionally, PQB standards may emerge from the PQC competition conducted by NIST. In this process, NIST rigorously tests, evaluates, and compares various PQC schemes to assess their efficiency, security, compatibility, and scalability. Currently, PQC standards could be finalized within a few years, while the development and deployment of quantum networks may take decades to mature.

C. AI-Driven Quantum Blockchain

Exploring the integration of quantum blockchain with AI opens up new research directions. By leveraging AI, routing and block propagation algorithms within quantum networks can be optimized, resulting in increased transaction throughput while preserving the decentralization of blockchain [186]. Moreover, AI models and algorithms can be applied to improve incentive mechanisms and quantum circuit structures for executing quantum consensus algorithms, thus boosting the security, efficiency, and scalability of quantum blockchain. This integration could also enhance the automation, efficiency, and adaptability of quantum blockchain applications, leading to innovations in decentralized networks, security protocols, and decision-making processes, thus strengthening the competitiveness of blockchain technology in Web 3.0.

D. Efficiency Optimization of Integrated Quantum Blockchain

Quantum blockchain faces significant efficiency challenges due to the complexity of its consensus mechanisms, as well as reliance on early-stage, costly quantum technologies and substantial energy demands.

First, current blockchain systems struggle with large transaction volumes, and the added computational complexity and heightened security requirements of quantum blockchain make scalability even more challenging, reducing efficiency as networks grow. Second, relying on expensive, early-stage quantum technology, including specialized hardware such as quantum processors, could aggregate resources. This concentration weakens decentralization, as only a few entities can afford to participate due to high costs. Lastly, while quantum computing offers immense power, its high operational energy costs make it essential to develop energy-efficient consensus protocols and algorithms to support the sustainable growth of quantum blockchain.

E. Decentralized Quantum Blockchain Applications for Web 3.0 Services

Decentralized digital identities allow users to manage their digital assets autonomously, securely, and transparently in Web 3.0. In quantum blockchain and PQBs, users can obtain verifiable, privacy-protected, and revocable digital identities without relying on third-party identity providers. Providing digital identities through QKD-based cryptography provides unconditionally secure data identity authentication and transmission. However, users need access to the quantum network to use applications in Web 3.0 through quantum digital identities. Therefore, the future research direction for quantum digital identity needs to be based on the improvement of quantum network throughput and coverage, so that users can protect their privacy and ownership through quantum communication in Web 3.0 according to different scenarios and needs.

By building Web 3.0 decentralized exchanges on quantum blockchain and PQB, traders can complete the transmission of transaction information under an absolutely secure communication link to prevent man-in-the-middle attacks. Quantum networks can be deployed between major exchanges to reduce the latency of information exchange between exchanges. In this way, the information gap between exchanges is dramatically reduced, thus avoiding some arbitrage activities due to network dominance. In addition, Web 3.0 allows its participants to borrow and lend without collateral, secured by quantum cryptography and PQC. Therefore, future research directions could focus on how to design highly liquid lending protocols in Web 3.0 through quantum and PQC to enhance the sustainability of the decentralized ecosystem.

F. Convergence of Classic-Quantum Networking and Computing

In Web 3.0, blockchain-based computational networks link diverse resources from classical and quantum systems to support DAPPs and services. By incorporating quantum blockchain, quantum computing and networks become integral to Web 3.0, enabling the allocation of computing and networking resources to optimize core technologies and enhance performance.

For instance, the convergence of classical and quantum networks allows Web 3.0 to leverage quantum computing for complex, large-scale tasks such as optimization, simulations, and ML. However, realizing these benefits presents significant challenges. While IBM's "Condor" quantum computer boasts over 1000 qubits [187], current quantum hardware remains limited, with high development and maintenance costs that hinder scalability and accessibility. Additionally, integrating quantum and classical systems brings interoperability challenges due to fundamental differences in data representation and processing.

Overcoming these barriers requires extensive research to establish robust protocols and algorithms for task allocation, scheduling, and synchronization across quantum and classical systems. Developing efficient methods for seamless convergence is also essential as Web 3.0 expands and DAPPs grow in complexity. These advancements are vital to fully harnessing the potential of quantum-classical hybrid systems, paving the way for scalable, secure, and efficient Web 3.0 ecosystems.

1) Lessons Learned: Transitioning blockchain systems to quantum and postquantum frameworks presents both challenges and opportunities. Ensuring a smooth migration path and prioritizing critical Web 3.0 services for quantum-resistant adaptation are essential steps. Standardization will play a vital role, as shared quantum and postquantum protocols can boost scalability and interoperability across digital ecosystems. Additionally, while integrating classical and quantum computing offers promising advantages, it requires tailored algorithms to manage this convergence effectively.

IX. CONCLUSION AND DISCUSSION

With the complementary strengths of quantum information technologies and blockchain, integrated quantum blockchain presents a promising approach for building distributed Web 3.0 infrastructures. In this article, we have explored quantum blockchain, providing a thorough review of key developments in Web 3.0. Our focus is on two main aspects: optimizing internal technologies and enhancing external performance. We have also outlined potential applications for quantum blockchain and provided guidance for implementation within the Web 3.0 ecosystem. Lastly, we have discussed significant challenges and research directions, which open new pathways for Web 3.0 advancement.

Quantum blockchain holds the potential to revolutionize data security, transaction speed, and decentralization. While quantum computing may pose risks to current cryptographic security, it also offers unique opportunities to enhance blockchain scalability and efficiency, addressing current transaction limitations. Quantum-resistant algorithms are expected to strengthen blockchain networks [188], while quantum-driven dApps could enable innovative applications in AI and complex simulations [189], [190]. Although in its early stages, quantum blockchain is anticipated to play an increasingly vital role in meeting Web 3.0's diverse needs. This work provides a comprehensive, systematic, and forward-looking introduction to quantum blockchain, supporting future research and applications across various sectors. ■

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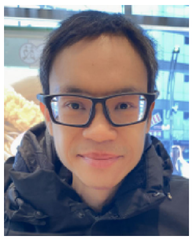

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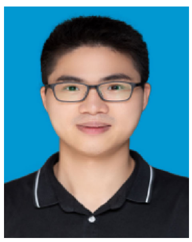

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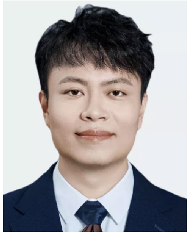

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